

Shopping by algorithm: How agentic AI deploys human heuristics as a surrogate consumer

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Abstract

Consumers increasingly delegate purchasing decisions to Large Language Models (LLMs) acting as surrogate consumers. Using “Tool-Lab,” an adaptation of information-board process tracing that places product attributes behind costly tool calls, we examine how marketing pricing cues (i.e., just-below pricing and promotional framing) influence AI shopping agents. Across eight commercially deployed LLMs from three providers, we trace pre-choice information acquisition. Under zero cost, pricing cues rarely mislead. Imposing acquisition costs under a vague goal prompt leads LLMs to omit diagnostic attributes required to compute unit price and choose suboptimal choices resembling human heuristics. Relative to a specific goal prompt that mainly preserves diagnostic search and choice optimality, a vague goal prompt under constraints creates a search-mediated vulnerability. This research demonstrates that marketing heuristics in delegated AI shopping are governed by storefront information architecture, not necessarily immutable LLM flaws.

Keywords: Artificial Intelligence, Large Language Models, Bounded Rationality, AI Agents

Introduction

The evaluation of market alternatives has historically relied on human judgment. Increasingly, however, consumers are delegating end-to-end purchasing tasks to autonomous Large Language Models (LLMs) deployed as surrogate consumers (Pandya; Aw et al., 2025; Hasselwander et al., 2026; Puntoni and Wertenbroch, 2024), that is, an external entity to whom a principal relinquishes decision authority to navigate the marketplace (Solomon, 1986). Major platforms have begun deploying agents that search, compare, and complete purchases on consumers’ behalf (Verdon; Sataua; Kahn). While surrogate consumers have traditionally been humans, delegating this role to AI agents represents a fundamental shift in who evaluates the market (Dholakia et al., 2025).

For decades, retail pricing strategies, such as just-below pricing (Lacetera et al., 2012; Stiving and Winer, 1997; Thomas and Morwitz, 2005) and promotional discount framing (Inman et al., 1990; Thaler, 1985), have been engineered to exploit human bounded rationality by triggering heuristic decision rules. Practitioners already advise retailers to structure pricing and product attributes so that AI agents can evaluate them (Cooper III). When the buyer is an autonomous AI agent, it remains an open question whether pricing cues built for human cognitive limitations continue to influence purchasing, and if so, how.

Research evaluating LLMs as economic actors has found them to produce both rational (Brand et al., 2023; Cillo and Rubera, 2025; Horton et al., 2023) and heuristic-like behavior (Binz and Schulz, 2023; Filandrianos et al., 2025; Hermann and Puntoni, 2025). Reconciling this duality has been proven difficult because the existing literature predominantly treats heuristic-like behavior as an inherent, static defect hardwired into LLMs during training. As a result, research has largely documented whether LLMs exhibit heuristic-like behavior, but not what produces it.

We propose that heuristic-like purchasing in delegated AI shopping can be a property of the information state that the AI agent holds at the moment it commits to a choice. This pre-choice information state can be jointly shaped by retailers and consumers. Retailers determine the cost of retrieving product attributes through platform design, and consumers determine, through the specificity of their delegating prompt, which

attributes the AI agent treats as diagnostic. The combination of costly retrieval and vague goal prompts lead AI agents to truncate their information search and choose suboptimally.

To test this claim, we introduce Tool-Lab, an experimental paradigm that adapts classical information-board process tracing (Johnson et al., 1989; Payne et al., 1993; Willemsen and Johnson, 2011) to multi-turn LLM interactions. Rather than presenting product attributes simultaneously within a single prompt, Tool-Lab embeds an LLM within an interactive environment where product attributes are hidden behind costly tool calls. To inspect an attribute, the LLM must execute a tool call (*inspect_cell*), which incurs an explicit acquisition cost. Tool-Lab records the observable sequence of information acquisitions prior to choice and allows researchers to measure search depth and composition directly. Therefore, Tool-Lab allows us to manipulate retrieval cost and prompt specificity while observing the search they produce. This makes the pathway from environmental constraint to final choice through information acquisition measurable, rather than inferred from choice outcomes alone.

Across two primary studies, several robustness tests, and two validation studies ($N = 16,400$ total sessions), we evaluate eight commercially deployed LLMs from three major providers (Google, OpenAI, and Anthropic). Each study establishes a full information baseline, then contrasts its choices with a constrained Tool-Lab environment where information acquisition incurs financial costs ($c = \$0$ vs. $\$10.00$ or $\$50.00$). Additionally, we manipulate the consumer prompt’s goal specificity (vague: “find the best deal,” vs. specific: “find the lowest price per ounce”).

The empirical results document two consistent findings. First, under full information and specific goal prompt, the evaluated LLMs make decisions that largely align with normatively optimal behavior. Second, when acquisition costs are imposed under a vague goal prompt, most LLMs decrease search depth, and the impact on choice optimality depends on whether the omitted attributes were diagnostic (i.e., they enter the unit price calculation). Because vague prompts lack a clear normative objective, LLMs abandon some diagnostic attributes (i.e., cents or product weight). Without the attributes that are required to calculate unit prices, the AI agents make suboptimal decisions that are consistent with human heuristic decision making. Validation experiments ($N = 5,200$) show that choice suboptimality stems from missing diagnostic attributes rather than arithmetic calculation failures for most LLMs.

This research makes three contributions to marketing theory of agentic AI shopping. First, it challenges the assumption that heuristic-like behavior in LLMs as shopping assistants is solely inherited from training data, by showing that susceptibility to marketing cues depends on the information state in which AI agents choose. Second, it identifies the principal’s delegation instruction as a determinant of whether marketing cues influence the AI agent’s choice. Third, it introduces Tool-Lab, a process-tracing method that makes the information acquisition of AI agents observable and allows researchers to study how AI agents arrive at decisions.

These findings offer implications for four stakeholders. Retailers should not rely on pricing cues designed for human shoppers when the buyer is an AI agent. Instead, they should compete by making diagnostic attributes readily available to AI shopping agents. Consumers should clearly state the criterion that defines a good choice (e.g., “lowest price per ounce”) rather than issuing vague requests such as “find the best deal.” AI agent developers should design agents to clarify vague objectives before searching and should audit the specific LLM they deploy for acquisition truncation. Regulators should consider extending disclosure rules, such as unit-price requirements, to the product information made available to AI agents.

Literature review

AI agents as surrogate consumers

Marketing scholarship has examined Generative Artificial Intelligence (GenAI) primarily through firm-side strategy (Prasanna and Kushwaha, 2025; Riandhi et al., 2025; Grewal et al., 2025; Kumar et al., 2026), yet the critical frontier in marketing theory centers on the autonomous AI agent itself (Aw et al., 2025; Dholakia et al., 2025; Puntoni and Wertenbroch, 2024). Behavioral research in this domain predominantly treats Large Language Models (LLMs) as passive “silicon samples” or synthetic respondent pools (Joerling, 2025; Sarstedt

et al., 2024) to emulate human survey responses and aid in marketing research (Arora et al., 2025; Li et al., 2025; Liu and Chen, 2026; Wang et al., 2026).

However, there is a growing use of AI agents by consumers as autonomous shopping assistants that perform end-to-end shopping (Hasselwander et al., 2026; Kim et al., 2023; Pandya; Google DeepMind, 2026). In marketing theory, delegating purchase authority to an external entity is a classic case of surrogate consumption (Solomon, 1986). A surrogate consumer is defined as an agent to whom a consumer relinquishes decision authority to evaluate, select, and acquire goods within the marketplace (Solomon, 1986). Whereas traditional surrogate consumers were humans, generative AI, in particular Large Language Models (LLMs), can act as an autonomous entity capable of querying APIs, exploring product catalogs, and committing financial transactions on a consumer’s behalf (Google DeepMind, 2026; Schick et al., 2023; Wang et al., 2024).

The rationality paradox

As LLMs assume the role of surrogate consumers, a behavioral paradox has emerged in the literature. LLMs neither conform strictly to classical economic assumptions of rationality (Thaler, 2000), nor do they predictably replicate human bounded rationality (Kahneman, 2003). On the one hand, an extensive body of research demonstrates that LLMs inherently possess strong analytical capabilities (Lee et al., 2026; Pandey and Singh, 2025). When prompted in standard discrete-choice environments, LLMs can function as highly capable simulated economic actors (Horton et al., 2023). They can evaluate expected values, respect downward-sloping demand curves, maintain budgetary compliance, and execute complex multi-attribute compensatory evaluations (Brand et al., 2023; Cillo and Rubera, 2025; Reisenbichler et al., 2026; ZHOU et al., 2026). These competencies embody analytical (“thinking”) AI, characterized by computational precision, logic, and rule-based data processing (Huang and Rust, 2021).

On the other hand, research in computational psychology documents that LLMs remain susceptible to classic cognitive anomalies (Binz and Schulz, 2023; Hermann and Puntoni, 2025). LLMs can show susceptibility to framing effects (Pescher, 2026), algorithmic impatience in intertemporal choices (Goli and Singh, 2024), and vulnerability to adversarial prompt perturbations (Xie, 2026). In product recommendation, promotional framing and sponsorship cues embedded in product text can distort LLM recommendations even when objective attributes remain the same (Filandrianos et al., 2025; Wadi and Ma, 2026b). Importantly, recent large-scale benchmarks evaluating cognitive biases demonstrate that this paradox is not resolved simply by increasing computational capability of the LLMs (Echterhoff et al., 2024; Knipper et al., 2025; Malberg et al., 2025). Thus, the emergence of heuristic behavior cannot be understood purely as a capability limitation. Instead, marketing scholarship must examine the contextual and task-level factors that govern when an LLM operates in an analytical versus a heuristic decision mode (Huang and Rust, 2021).

Marketing pricing cues

While much of the computer science literature examines cognitive biases through abstract logic puzzles, such as the Linda problem or the Wason selection task (Binz and Schulz, 2023; Knipper et al., 2025), marketing phenomena introduce a fundamentally different evaluative dynamic. In consumer purchasing, pricing tactics (e.g., just-below pricing and promotional discount framing) do not operate solely as abstract semantic frames. Rather, these marketing cues directly compete with exact utility calculations (Stiving and Winer, 1997; Thomas and Morwitz, 2005).

Just-below pricing (e.g., pricing an item at \$4.99 rather than \$5.00) exploits a heuristic wherein decision-makers focus disproportionately on the leftmost integer while ignoring fractional cents (Bizer and Schindler, 2005; Thomas and Morwitz, 2005; Kraft and Rao, 2026; Lacetera et al., 2012).

Similarly, promotional discount framing (e.g., “20% off”) exploits consumer transaction utility (Thaler, 1985) by signaling a bargain to the consumer (Inman et al., 1990). If an AI shopping agent attends solely to the whole dollar price or the promotional cue while ignoring cents or package volume, it would select an economically inferior product.

Storefront information architecture

The marketing and computer science literatures predominantly treat heuristic susceptibility in LLMs as a static, immutable defect hardwired during training (Echterhoff et al., 2024; Knipper et al., 2025; Malberg et al., 2025). According to this view, an LLM defaults to choices resembling human heuristics because its autoregressive parameters reflect the bounded rationality deeply embedded within the human text.

However, this remains an untested and largely untestable claim. Because modern LLMs are trained on uncensored, proprietary web-scale corpora spanning trillions of tokens, empirically verifying whether a suboptimal choice stems from heuristics embedded directly into its parameters is virtually impossible. Despite the absence of direct empirical verification, this “innate defect” account has operated as an implicit assumption across computer science research evaluating anomalies in large language models.

This assumption has persisted largely due to a prevailing methodological constraint. When an AI shopping agent is presented with fully enumerated product profiles in a single prompt and selects an inferior option, the researcher observes only the terminal choice (Johnson et al., 2008). Recent audits of AI shopping agents, for example, document position biases and choices that vary across LLMs from final choices alone (Allouah et al., 2026; Pezeshkpour and Hruschka, 2024; Wadi and Ma, 2026a; Liu et al., 2024; Yin et al., 2026). This outcome-based data cannot reveal whether the LLM calculated unit values but over-weighted a marketing cue, or whether its informational state was configured such that normative calculation was never attempted.

We propose an alternative account. Choices that resemble human heuristics in AI agents are not solely innate training defects. Rather, they are a product of the interplay between the information architecture of the storefront and the delegating prompts used by consumers.

Conceptual development

Normative and heuristic objectives

In economic decision-making, an agent evaluates market alternatives against a specific objective function. Depending on the decision context, this objective would optimize nominal expenditure, brand equity, perceived quality, or price per unit of volume. In the consumption of packaged goods, marketing theory and consumer economics establish that the normative objective is unit-price minimization, that is, selecting the alternative j that minimizes financial expenditure per unit of physical quantity (Miyazaki et al., 2000; Russo, 1977):

$$\min_{j \in \mathcal{J}} \left(\frac{\text{Price}_j}{\text{Weight}_j} \right)$$

It is strictly against this normative benchmark that behavioral economics considers choice driven by pricing cues, such as just-below pricing and promotional discount framing, economically suboptimal, or refers to them as biases.

Consider just-below pricing. If an agent chooses a 10-ounce package priced at \$4.99 over an available 100-ounce package priced at \$5.00, the decision is optimal under a nominal price-minimization objective ($\min_{j \in \mathcal{J}} \text{Price}_j$), but suboptimal under the normative unit-price objective ($\min_{j \in \mathcal{J}} \frac{\text{Price}_j}{\text{Weight}_j}$). In human decision research, when an individual shifts from the normative objective to a simpler objective, this substitution is defined as a heuristic (Gigerenzer and Gaissmaier, 2011; Shah and Oppenheimer, 2008).

Evaluating promotional discounts similarly follows. Computing the normative objective of a discounted item requires evaluating a multi-parameter objective function:

$$\min_{j \in \mathcal{J}} \text{Unit Price}_j = \min_{j \in \mathcal{J}} \frac{\text{List Price}_j \times \left(1 - \frac{\text{Discount Percentage}_j}{100} \right)}{\text{Weight}_j}$$

Because computing this objective requires multi-step arithmetic, decision-makers operating under constraints frequently substitute the normative objective with a computationally simpler objective, such as maximizing the discount percentage or minimizing the discounted list price. In marketing theory, this objective substitution is called transaction utility (Inman et al., 1990; Thaler, 1985).

Following this framing, we define the normative objective for consumer packaged goods as unit-price minimization and classify any systematic departure from the normative objective as choice suboptimality. Heuristic then becomes a resource-conserving departure from full evaluation of the normative objective, either by substituting a simpler objective or by evaluating the normative objective on a reduced set of attributes (Shah and Oppenheimer, 2008).

Product attributes can be classified by their role in the normative objective calculation. We define the subset of attributes that enter the unit-price equation as diagnostic (i.e., list price, whole dollars, cents, discount percentages, and weight), and the remaining attributes as non-diagnostic (e.g., packaging, country of origin, calories).

Acquisition Costs

In human judgment, information acquisition is costly. Human decision-makers face both external constraints (e.g., time pressure, fragmented shelf displays) and internal cognitive limitations (e.g., working memory capacity, computational fatigue; Payne et al., 1993; Shugan, 1980).

Similarly, an LLM operating as a surrogate consumer incurs both external and internal acquisition costs. External acquisition costs arise from storefront architecture and environment design, including the structural placement and prominence of attributes within the prompt or the structure of API payloads. Internal acquisition costs reflect computational resource allocation and provider-level LLM tuning. These include the LLM’s allocated reasoning effort and its propensity to execute multi-turn tool calls before terminating search. Foundation model providers calibrate this trade-off during instruction-tuning and Reinforcement Learning from Human Feedback (RLHF) to optimize inference speed and API operating costs against output accuracy. If an AI agent were to exhaustively inspect every latent attribute via unbounded tool calls, task execution would become prohibitively slow and expensive.

Throughout this research, we use the term *acquisition cost* to refer to both the internal and external constraints that an LLM operates within.

Goal specificity

LLMs are instruction-tuned to deliver contextually coherent and relevant responses to users’ prompts. In a delegation relationship, however, task performance strongly depends on the goal specificity communicated by the delegating principal (Kleingeld et al., 2011; Locke and Latham, 2002). In accordance with goal-setting theory (Latham, 2023), high goal specificity directs the agent’s effort to goal-relevant information.

A *specific goal prompt* explicitly defines the normative objective function to be optimized (e.g., “*Find the coffee with the lowest price per ounce*”), which defines the diagnostic attribute set (i.e., any attribute that determines price and weight).

Conversely, a *vague goal prompt*, such as the colloquial “*Find the best deal*”, leaves the objective open to multiple valid interpretations (e.g., price per weight or price). Therefore, such a prompt leaves objective specification partly to the LLM.

Under high acquisition costs, a vague goal prompt can reduce diagnostic acquisition through two routes. First, the LLM can substitute the normative objective (price per weight) with a simpler one (e.g., price), which shrinks the diagnostic set from attributes that determine price and weight to attributes that determine price alone. Second, the LLM can omit attributes whose expected value of information is low relative to their acquisition cost, such as cents, whose range (\$0.00 to \$0.99) is small relative to whole dollars (Web Appendix H). Both routes reduce the total acquisition cost of completing the task, but both lead to suboptimal choices

when the omitted attributes determine the ranking of alternatives. We do not seek to distinguish between these two routes, as both yield the hypothesized pathway¹:

H1. A vague goal prompt negatively moderates the relationship between acquisition cost and diagnostic acquisition.

Because computing the normative objective requires the complete set of diagnostic attributes, the diagnostic information retrieved dictates whether the LLM has the necessary information for normative objective optimization.

H2. Diagnostic acquisition positively influences choice optimality.

Integrating these stages establishes a conditional indirect process. Under a vague goal prompt, acquisition costs induce objective substitution and diagnostic acquisition truncation, leading to choice suboptimality.

H3. The indirect effect of acquisition cost on choice optimality via diagnostic acquisition is moderated by prompt specificity.

Insert Figure 1 about here

We first present the Tool-Lab paradigm, which allows us to test our hypotheses. We then report two empirical studies examining just-below pricing (Study 1) and promotional discount framing (Study 2). Each study begins with a baseline where all attributes are visible, followed by a Tool-Lab experiment where attributes are hidden behind costly tool calls.

The Tool-Lab paradigm

To test how acquisition costs govern decision-making in delegated AI purchasing, empirical research requires a method that moves beyond choice outcomes. In human decision research, process-tracing methods such as Mouselab (Johnson et al., 1989; Payne et al., 1993) externalize internal decision-making steps by placing product attributes behind closed display boxes on a computer screen. By tracking which boxes human participants uncover, researchers observe information acquisition directly.

We adapt this paradigm to Large Language Models (LLMs) acting as consumption decision makers through an experimental paradigm we term Tool-Lab. Rather than presenting an LLM with all product attributes within a single prompt, Tool-Lab embeds the LLM within an interactive, stateful software environment where product data are hidden behind costly tool calls.

In Tool-Lab, product alternatives and their corresponding attributes are structured as an $M \times N$ matrix of hidden cells. The AI agent interacts with this environment using tool calls (function calling). The environment exposes two tools, namely, *inspect_cell* and *submit_choice*². To reveal an attribute’s value, the LLM must execute a tool call specifying the alternative and the attribute (e.g., *inspect_cell(option_id="coffee_b", attribute_id="weight")*)³. In response, the environment returns a structured JSON payload containing the revealed value, the cost incurred for the tool call, and the cumulative cost accrued during the session. When the LLM stops searching, it terminates the search sequence by invoking a separate tool, named *submit_choice* (e.g., *submit_choice(option_id="coffee_b")*). Invoking this tool concludes the session and commits the final purchase.

To operationalize the acquisition cost, Tool-Lab attaches an explicit financial cost to each invocation of *inspect_cell*, which is charged directly to the delegating consumer. *submit_choice* carries zero cost. By manipulating this cost across conditions, we observe how the AI agent adjusts its information acquisition behavior under varying degrees of constraint (Figure 2).

Insert Figure 2 about here

¹Distinguishing between these two routes across LLMs and pricing cues is beyond the scope of this research, although we illustrate how Tool-Lab can separate them in Web Appendix E

²During each session, the LLM can execute only one tool call per turn.

³To prevent the letters “a” or “b” in coffee_a and coffee_b affecting the choice, each session randomizes the *option_id*.

By recording the sequence of information acquisitions executed by the LLM prior to choice, we quantify total acquisition, the distribution of acquisition across diagnostic (vs. non-diagnostic) attributes, and the final choice. This allows us to isolate whether a suboptimal choice occurred under complete diagnostic information (a utilization failure) or under incomplete diagnostic information (an acquisition failure).

Tool-Lab can also record the reasoning traces that LLMs produce between tool calls (Yao et al., 2022). This is conditional on LLMs having reasoning capabilities and providers making reasoning traces accessible. See Web Appendix E for an example of how we used the pre-tool reasoning traces to classify the inferred objectives expressed by LLMs under vague goal prompt.

Stochastic evaluation. Evaluating LLMs in multi-turn tool environments requires addressing the stochastic nature of autoregressive generation. In standard commercial deployments, LLMs sample tokens from probability distributions, meaning identical prompts can generate heterogeneous search trajectories and final decisions.

To account for this variance, Tool-Lab implements a Monte Carlo sampling protocol (Wadi and Fredette, 2025; Robert et al., 1999; Holtzman et al., 2019). Rather than evaluating a single deterministic pass, each experimental cell is evaluated across multiple independent simulated sessions per LLM. Every session is initiated as an isolated API conversation with zero state retention between trials. Within each session, the presentation order of alternative options and attribute rows is randomized to prevent position confounds. All interactions are executed programmatically via official client libraries under default provider decoding configurations, and all tool responses are validated against strict JSON schemas (detailed specifications and snapshot identifiers are provided in Web Appendix I).

LLMs tested. Unless noted otherwise, the experiments test eight commercially deployed LLMs from three major providers: Gemini 3.5 Flash Lite (Flash Lite), Gemini 3.8 Flash (Flash), and Gemini 3.1 Pro (Pro) from Google, GPT 5.6 Luna (Luna), GPT 5.6 Terra (Terra), and GPT 5.6 Sol (Sol) from OpenAI, and Claude Sonnet 5 (Sonnet) and Opus 5 (Opus) from Anthropic (shorthand names in parentheses). To avoid repetition, we refer to these LLMs using their shorthand name.

Study 1

Baseline

We first test LLM choice when diagnostic attributes (i.e., dollars, cents, and weight) are fully available without any external constraints. We constructed a coffee choice experiment forcing trade-offs between brand, product weight, and price. We utilized four best-selling Amazon.com brands (*McCafé*, *Maxwell House*, *Starbucks*, *Folgers*). To investigate the relative effect of the cent versus the dollar value of the price on choice, we independently drew dollars and cents from the real-word distribution of coffee prices. We generated 1,014 non-dominated choice sets to ensure no alternative was superior in all attributes, which would push the LLMs to make genuine economic trade-offs.

We queried the eight LLMs as well as a normative objective optimizer that always chooses the alternative with the lowest price per weight. We used a standardized prompt delegating the LLM as a surrogate shopping assistant (e.g., “You are buying ground coffee. Which option would you choose?”). The presentation positions of the alternatives were fully randomized to control for position effects. Details on choice set generation and prompt structure are provided in Web Appendix B.

We estimate an independent conditional logit for each LLM and the normative objective optimizer separately, assuming a random utility framework (McFadden, 1972; Train, 2009, specification in Web Appendix B). Table W1 shows the results.

Normative price sensitivity requires that a cent in dollar units be weighted at the same rate as a dollar. To test whether the LLMs weight the cents at the normative rate, we estimate separate marginal utilities for whole dollars ($\beta_{\text{Dollars } (\$)}$) and for fractional cents converted to dollar units ($\text{Cents } (\$) = \frac{\text{Cents}}{100}$), and evaluate their ratio, $\beta_{\text{Cents } (\$)} / \beta_{\text{Dollars } (\$)}$, against the normative benchmark of one. Confidence intervals are exact, obtained by Fieller’s theorem (Fieller, 1954) from the estimated coefficients.

The ratio of cent-to-dollar sensitivity ranges from 0.55 to 1.12, and seven out of eight LLMs’ 95% confidence interval contains 1.00 (Table W2). Under full information, we find that Flash discounts cents with respect to whole-dollars. For the remaining LLMs, we find no evidence that they systematically discount or favor the cent component of a price relative to the whole-dollar component.

Tool-Lab

While the baseline experiment shows that LLMs evaluate cents at a similar rate to dollars or less, under full attribute information, it cannot trace what happens when the information environment is constrained, and how acquisition costs affect LLM choice. We deploy Tool-Lab to explore this and test our hypotheses.

Stimuli and sample

We instruct the LLMs to select an instant coffee for a human consumer. The stimulus set consists of two alternatives designed as a choice trap to isolate the just-below pricing effect. *Coffee A*, features a lower whole-dollar price (\$4.99 for 10 oz), but yields a suboptimal unit-price of 0.499 \$/oz. *Coffee B*, the optimal choice, features a higher whole-dollar price (\$5.00 for 11 oz), but has the superior unit price at 0.455 \$/oz. A normative objective optimizer would thus choose Coffee B.

To force a tradeoff between the dollar and cent components of the price, We separate the price into two hidden attributes, *price_dollars* (e.g., “\$4”) and *price_cents* (e.g., “99 cents”). To calculate the unit price ($\frac{\text{price_dollars} + \text{price_cents}}{\text{weight}}$), the LLM must expend resources to reveal three diagnostic attributes for each alternative (6 diagnostic attributes total). To increase ecological validity of the products and to obscure the true purpose of the experiment, we also include four non-diagnostic attributes (*origin*, *roast_date*, *calories*, and *packaging*). The values for these attributes are held constant for both alternatives (i.e., “Colombia”, “2 weeks ago”, “3” “Foil bag”) to prevent any confounds.

We implement a 2 (Acquisition cost: \$0.00 vs. \$10.00 per *inspect_cell* invocation) $\times$ 2 (Goal specificity: vague vs. specific goal prompt) between-sessions design. To manipulate goal specificity, we vary the evaluative instruction embedded in the prompt. In the vague goal prompt condition, the LLM receives a vague, colloquial purchasing prompt: “You need to find the **best deal** on instant coffee.” In the specific goal prompt condition, this instruction is replaced with an specific, normative objective: “You need to find the coffee with the **lowest price per ounce**.” All other elements of the prompt, the product stimuli, and the tool environment are held constant across conditions. We evaluate the eight LLMs. The presentation order of the alternatives and attribute rows is randomized across sessions. In total, Study 1 Tool-Lab comprises 3,200 unique simulated sessions (8 LLMs $\times$ 2 acquisition costs $\times$ 2 goal specificity conditions $\times$ 100 iterations).

Procedure and measures

Each session begins with a system prompt establishing the LLM role: “You have been delegated by a consumer to make a purchase decision. Your goal is to choose an instant coffee for the user. After you choose a coffee, the user would purchase 5 bags of your chosen coffee.”⁴ The prompt defines the tool mechanics and cost constraint: “You can open the cells to get information about the product using the *inspect_cell* tool, or you can stop searching and submit your choice with the *submit_choice* tool. Every time you use the *inspect_cell* tool, the user loses {*inspect_tool_cost*}. *submit_choice* has no cost.” This is followed by the goal specificity manipulation (“You need to find the best deal on instant coffee” vs. “You need to find the coffee with the lowest price per ounce”), concluding with the choice query: “Which coffee would you choose for the user?” See Table W3 for the prompts. Figure 2 demonstrates the Tool-Lab setup for the \$10 per *inspect_cell* condition.

Choice optimality is the final product selected via the *submit_choice* tool, coded as a binary variable (1 = the optimal choice; 0 = the suboptimal choice). *Diagnostic acquisition* is the count of how many of the six diagnostic attributes (*price_dollars*, *price_cents*, and *weight*) the LLM inspects before submitting its choice.

⁴We scale the purchase volume to five bags to ensure that the potential economic surplus of identifying the optimal product mathematically justifies the cumulative tool costs.

Results

Aggregated across all LLMs tested (Table 1; per LLM descriptive in Table W4), under the specific goal prompt at zero cost, diagnostic acquisition was complete across all attributes ($M = 6.00, SD = 0.00$), and choice optimality was high ($M = 0.94, SD = 0.16$). Imposing the \$10.00 acquisition cost produced a small reduction in diagnostic acquisition ($M = 5.72, SD = 0.59$) and choice optimality ($M = 0.83, SD = 0.30$). Attribute-level acquisitions show that weight ($M = 2.00, SD = 0.00$) and dollars ($M = 2.00, SD = 0.00$) remained at ceiling. The reduction was driven exclusively by the omission of cents ($M = 1.72, SD = 0.59$).

Under the vague goal prompt at zero acquisition cost, diagnostic acquisition ($M = 5.86, SD = 0.24$) and choice optimality ($M = 0.89, SD = 0.19$) were high, with high acquisitions for dollars ($M = 2.00, SD = 0.00$), cents ($M = 1.97, SD = 0.09$), and weight ($M = 1.90, SD = 0.18$). However, imposing the \$10.00 acquisition cost under the vague goal prompt produced a substantial reduction in diagnostic acquisition ($M = 3.81, SD = 1.30$) and choice optimality ($M = 0.29, SD = 0.28$). Examining individual attributes shows that while acquisitions for dollars remained high ($M = 1.94, SD = 0.09$), the LLMs reduced both cents ($M = 0.77, SD = 0.78$) and weight ($M = 1.10, SD = 0.64$).

These aggregate patterns show that under a specific goal prompt, acquisition costs lead the LLMs to maintain weight and dollars but ignore cents. Under a vague goal prompt, however, acquisition costs lead the LLMs to largely maintain dollars but ignore weight and cents.

Insert Table 1 about here

Moderated mediation

To test our hypotheses (H1–H3), we examine the structural pathways connecting acquisition cost, prompt specificity, diagnostic acquisition, and choice optimality. Unlike behavioral experiments with human participants who typically each provide a single observation, our empirical setting evaluates a sample of LLMs across repeated, independent Monte Carlo sessions. To test whether the hypothesized effects generalize across the evaluated LLMs while allowing baseline acquisition propensities and structural slopes to vary across each LLM, we implement a Bayesian multilevel moderated mediation in Stan via *brms* (Bürkner, 2017).

The first stage specifies diagnostic acquisition as a function of acquisition cost, prompt specificity (vague vs. specific goal prompt), and their interaction (H1). The second stage specifies choice optimality as an additive function of diagnostic acquisition (H2), acquisition cost, and prompt specificity. To avoid the identification anomalies and non-linear scale distortions of logistic mediation, the second stage utilizes a linear probability specification, which permits an exact additive decomposition of conditional indirect effects (Breen et al., 2013) and formal evaluation of the Index of Moderated Mediation ($a_3 \times b_1$; H3). Econometric equations, prior specifications, and Markov chain diagnostics are reported in Web Appendix D.

Insert Table 2 about here

The empirical estimates are consistent with the hypothesized structural pathways (Table 2). In the first stage, prompt specificity moderates the relationship between acquisition cost and diagnostic acquisition ($a_3 = -1.727, SD = 0.467, 95\% \text{ CI } [-2.641, -0.765]$), supporting H1. Specifically, the negative effect of acquisition cost on diagnostic acquisition is significant under a vague goal prompt ($a_1 + a_3 = -1.997, SD = 0.487, 95\% \text{ CI } [-2.952, -1.003]$), compared to the specific goal prompt ($a_1 = -0.270, SD = 0.217, 95\% \text{ CI } [-0.698, 0.161]$).

In the second stage, diagnostic acquisition is positively associated with choice optimality ($b_1 = 0.263, SD = 0.034, 95\% \text{ CI } [0.196, 0.329]$), supporting H2. Consistent with H3, prompt specificity moderates the indirect pathway from acquisition cost to choice optimality via diagnostic acquisition, as evidenced by the significant Index of Moderated Mediation ($\text{IMM} = -0.451, SD = 0.128, 95\% \text{ CI } [-0.704, -0.194]$). Specifically, the conditional indirect effect is negative under a vague goal prompt (indirect effect = $-0.522, SD = 0.132, 95\% \text{ CI } [-0.783, -0.256]$), but overlaps zero under a specific goal prompt (indirect effect = $-0.070, SD = 0.057, 95\% \text{ CI } [-0.184, 0.044]$). Finally, the direct effect of acquisition cost on choice optimality is not credibly different from zero ($c' = -0.100, SD = 0.076, 95\% \text{ CI } [-0.251, 0.051]$), indicating that the observed decrease in choice optimality under acquisition cost operates primarily through the truncation of diagnostic acquisition.

Disaggregated analyses indicate that this indirect pathway is broadly consistent across the evaluated LLMs, with one exception. Luna maintained diagnostic acquisition under vague goal prompt with an IMM that overlapped zero ($\text{IMM} = -0.063$, 95% CI $[-0.158, 0.033]$). LLM-specific structural path estimates and conditional indirect effects are reported in Table W5 and Table W6.

Prompt variation

A natural concern with LLM evaluations is that observed decisions may reflect idiosyncratic prompt wording rather than the causal pathway that we claimed. Holding choice stimuli and tool mechanics invariant, we test three prompt variations (Table W7): the *Original* prompt; *Sentence reorder*, which moves the goal instruction earlier in the prompt to test for recency effects; and *Cost paraphrase*, which replaces the loss framing with fee framing to test for lexical loss confounds.

Crossing these 3 prompt variations with 2 acquisition costs (\$0 vs. \$10) and 2 prompt specificities (vague vs. specific goal prompt) yields 12 cells. Because testing the full design of Study 1 Tool-Lab for all eight LLMs across 12 cells would be combinatorially prohibitive and difficult to report, we focus on the Google Gemini family ($n = 100$; $N = 3,600$ sessions total, including 2,400 new sessions). Gemini LLMs exhibited wide behavioral variance in Study 1 Tool Lab, under vague prompt, from a small cost-induced truncation in Flash Lite ($a_3 = -0.48$) to a pronounced truncation in Pro ($a_3 = -3.23$; Table W5).

To evaluate the structural pathways within each condition, we estimated the two-stage moderated mediation system separately for each LLM $\times$ prompt variation subset ($n = 400$) using Bayesian simultaneous linear equations in *brms*, applying the linear probability specification and regularizing priors established in the primary analysis (Web Appendix D).

Across prompt variations, the structural estimates replicate the primary findings for two of the three LLMs tested (Table W8). For Pro and Flash, prompt specificity credibly moderates the relationship between acquisition cost and diagnostic acquisition across all three phrasing variations (H1), with a_3 ranging from -2.63 to -3.20 for Pro and from -2.06 to -2.56 for Flash (all 95% CIs exclude zero). Diagnostic acquisition remains positively associated with choice optimality in every condition (H2), with b_1 estimates ranging from 0.180 to 0.278 (all 95% CIs exclude zero). The Index of Moderated Mediation remains credibly negative across all phrasing variations for both LLMs (H3), ranging from -0.474 to -0.706 for Pro and from -0.488 to -0.572 for Flash.

For Flash Lite, the interaction (a_3) and the Index of Moderated Mediation (IMM) are credibly negative under the original prompt ($a_3 = -0.441$, 95% CI $[-0.698, -0.186]$; $\text{IMM} = -0.133$, 95% CI $[-0.219, -0.054]$) and the cost paraphrase ($a_3 = -0.519$, 95% CI $[-0.750, -0.289]$; $\text{IMM} = -0.140$, 95% CI $[-0.218, -0.072]$), but overlap zero under sentence reordering ($a_3 = -0.120$, 95% CI $[-0.375, 0.140]$; $\text{IMM} = -0.029$, 95% CI $[-0.093, 0.034]$). Diagnostic acquisition continues to positively predict choice optimality for Flash Lite across all three variations ($b_1 = 0.241$ to 0.302 , all 95% CIs exclude zero).

The results show that the hypothesized pathways are robust across phrasings for two LLMs with pronounced baseline truncation (Pro and Flash). For Flash Lite, sentence reordering acts as a boundary condition. Placing the objective earlier in the prompt reduced the first-stage acquisition truncation to non-significance ($a_3 = -0.120$), and accordingly attenuated the indirect effect ($\text{IMM} = -0.029$). Diagnostic acquisition continued to credibly predict choice optimality in every cell ($b_1 > 0$).

Price partitioning

A potential concern with the Tool-Lab environment in Study 1 is that the effect of acquisition cost on choice optimality might be an artifact of partitioning the price into two separate fields (*price_dollars* and *price_cents*), which is not how prices are naturally represented in retail environments. If the observed vulnerability to just-below pricing is merely a byproduct of this artificial separation, choice suboptimality would reflect a measurement artifact of the Tool-Lab paradigm rather than a genuine behavioral pathway under information acquisition constraints.

To test this, we conducted an experiment where price is acquired as a single unified *price* attribute. We implemented a 2 (Acquisition cost: \$0.00 vs. \$10.00) $\times$ 2 (Goal specificity: vague vs. specific) between-sessions

design. We evaluated three LLMs from Google (Flash Lite, Flash, and Pro). Each cell was evaluated across 100 independent sessions, yielding 1,200 unique sessions in total. All other stimulus characteristics, non-diagnostic attributes (four cells per alternative; eight cells total), and Tool-Lab interaction mechanics were identical to Study 1 Tool-Lab.

The results confirm that the pathway shown in Study 1 Tool-Lab is not an artifact of the price partitioning. Rather, consolidating price components shifts acquisition truncation from cents to weight and replicates the hypothesized structural pathways (H1–H3). Under the specific goal prompt (“lowest price per ounce”), acquisition costs have zero effect on diagnostic acquisitions and optimality (Table W9 and Table W10). In contrast, under the vague goal prompt, acquisition costs a large reduction in diagnostic acquisitions and choice optimality. The LLMs inspected the *price* cells for both alternatives in most session under cost ($M = 1.96$ to 2.00), but reduced acquisitions of *weight* considerably ($M = 0.56$ to 1.71).

Bayesian simultaneous equations estimation confirms that this choice degradation is driven by diagnostic acquisition truncation (Table W11). In the first stage, prompt specificity credibly moderated the effect of acquisition cost on diagnostic acquisition across all three LLMs, supporting H1. The interaction parameter (a_3) was negative for Flash Lite ($a_3 = -0.721$, 95% CI $[-0.981, -0.462]$), Flash ($a_3 = -0.332$, 95% CI $[-0.489, -0.175]$), and Pro ($a_3 = -1.320$, 95% CI $[-1.510, -1.120]$). In the second stage, diagnostic acquisition strongly and positively predicted choice optimality for all three LLMs, supporting H2. The marginal effect (b_1) ranged from 0.448 to 0.495 (all 95% CIs exclude zero). Supporting H3, the Index of Moderated Mediation ($IMM = a_3 \times b_1$) was credibly negative for all three LLMs: Flash Lite ($IMM = -0.357$, 95% CI $[-0.486, -0.228]$), Flash ($IMM = -0.149$, 95% CI $[-0.219, -0.078]$), and Pro ($IMM = -0.622$, 95% CI $[-0.716, -0.526]$).

These results demonstrate that the failure to optimize under constraint does not depend on partitioning prices into separate whole-dollar and cent fields. When price components are unified, acquisition cost under vague goal prompt leads the LLMs to truncate weight acquisitions, which leads to suboptimal choices.

Sensitivity to purchase volume

In Study 1 Tool-Lab, Pro acquired diagnostic attributes incompletely under the vague goal prompt even when acquisition was free (cents: $M = 1.75$; weight: $M = 1.68$; Table W4). Our conceptual development identifies two routes through which a vague goal prompt can reduce diagnostic acquisition: objective substitution and truncation of low-value attributes. The two routes respond differently when the benefit of information increases. If Pro retains the normative objective under the vague goal prompt but omits attributes because their value of information is low, increasing the number of bags purchased (purchase volume) should not decrease acquisition, because the surplus from identifying the optimal alternative grows with volume.

If instead Pro substitutes the normative objective with a simpler one under the vague goal prompt, purchase volume can shift which objective Pro uses, and acquisition should follow the diagnostic set of that objective rather than that of the normative objective. Weight acquisition is especially informative because weight is a diagnostic attribute under unit-price minimization (Web Appendix H), but non-diagnostic under price minimization.

To test which route governs the observed pattern, we implemented a 3 (LLMs: Flash Lite, Flash, and Pro) $\times$ 4 (Purchase volume: 1, 5, 50, and 250 bags) between-sessions design under the vague goal prompt at zero acquisition cost. Each cell was evaluated across 100 independent sessions, yielding 1,200 unique sessions in total. All other stimulus characteristics, attribute structures (six partitioned cells for price components and weight), and Tool-Lab mechanics were identical to Study 1 Tool-Lab.

The results show different patterns for the LLMs tested. Flash remained at ceiling ($M = 6.00$; optimality = 1.00), but scaling from 1 to 250 bags reduces diagnostic acquisitions for Flash Lite and Pro (Figure W1; Table W12). Pro reduced diagnostic acquisition from $M = 5.55$ to 4.22 (cents: 1.81 to 1.30; weight: 1.74 to 0.92) and optimality from .80 to .35. Flash Lite retained cents ($M \approx 2.00$) but reduced weight acquisitions ($M = 1.82$ to 1.30), diagnostic acquisition ($M = 5.77$ to 5.30) and optimality from .62 to .37. Scaling purchase volume thus increases suboptimality for Flash Lite and Pro.

To evaluate what objective the LLMs adopted⁵, an independent judge LLM classified the first reasoning trace emitted before any tool execution into unit price (*price_per_weight*), absolute price (*price*), or other (Web Appendix E). To validate this classification, an independent human evaluator blindly coded a stratified subsample of 120 traces. Agreement between the human rater and the judge LLM was high across all three categories (89.2% agreement, Cohen’s $\kappa = .78$; Table W26).

Based on the classification, Flash adopted a unit-price minimization objective (*price_per_weight*) in 90.8% to 94.0% of sessions and adopted an absolute price minimization objective in at most 4.0% of sessions (Table W13; Figure W1, Panel A). In contrast, the objective Pro adopted shifted with purchase volume. At one bag, 62.0% of Pro sessions started by articulating a unit-price objective, with 38.0% articulating an absolute price objective. At 50 and 250 bags, absolute price objective increased monotonically (83.0% and 84.0%, respectively; Table W13).

Under an absolute price objective, cents and product weight become functionally non-diagnostic because Coffee A’s whole-dollar price (\$4.xx) is lower than Coffee B’s (\$5.00). Therefore, the AI agent prematurely terminates search without inspecting cents or weight.

To examine whether the adopted objective, rather than the purchase quantity itself, is associated with search truncation and suboptimal choice, we evaluated the contrast between sessions adopting an absolute price objective and those adopting a unit-price objective, holding purchase volume fixed (Table W14; Web Appendix E).

Within every volume condition, Pro sessions adopting an absolute price objective acquired significantly fewer diagnostic attributes (differences ranging from -1.10 to -1.95 cells; all 95% bootstrap CIs exclude zero) and were 42 to 64 percentage points less likely to select the optimal alternative (Table W14).

Study 2

Study 2 investigates our conceptual framework under another marketing cue: promotional discount framing. Evaluating a discounted product requires multi-attribute arithmetics. An LLM must apply a percentage discount to a base list price to determine net expenditure, and then divide that net cost by product weight to derive unit value $\left[\frac{\text{List Price} \times (1 - \text{Discount})}{\text{Weight}} \right]$.

Baseline

We first examine how LLMs process promotional framing when all attributes are freely accessible. Discounted alternatives display the list price, product weight, and a promotional discount percentage (manipulated from 5% to 40% in 5% increments). Thus, optimizing the normative objective requires the LLM to compute its unit price.

Following Study 1, we constructed 1,149 Pareto-filtered, non-dominated choice sets of four ground coffee alternatives spanning the same four Amazon brands (*McCafé*, *Maxwell House*, *Starbucks*, *Folgers*). A random 50% of alternatives were assigned a promotional discount. Discount percentages and list prices were independently drawn. We evaluated the eight LLMs from Study 1. Full choice set generation algorithms, prompt templates, and econometric specifications are detailed in Web Appendix C.

We estimate an independent conditional logit model for each LLM independently as well as a normative objective optimizer that always chooses the alternative with the lowest unit price. The random utility framework detailed in Web Appendix C. An alternative’s net price is $\text{List Price} \times (1 - \text{Discount})$. A specification in which the LLM correctly synthesizes the discount implies a negative coefficient on base list price and a positive coefficient on the discount amount (in dollars). The estimation results across all eight LLMs are reported in Table W15.

⁵We use *adopted objective* to contrast the objective an LLM works toward under a vague goal prompt with the objective stated in the specific goal prompt.

Across all eight LLMs, base list price enters negatively ($\beta_{LP} = -1.06$ to $-3.08, p < .001$) and discount amount (in dollars) enters positively ($\beta_{DA} = 0.76$ to $3.08, p < .001$). Furthermore, product weight is a significant, positive predictor of choice for every LLM ($\beta_W = 0.78$ to $1.79, p < .001$).

To evaluate whether the LLMs treat a dollar saved through discounts at the same rate as a dollar spent through list prices, we examine the ratio of marginal utility for discount savings relative to list price disutility ($\beta_{\text{Discount Amount}}/|\beta_{\text{List Price}}|$; Table W16), with exact 95% confidence intervals derived via Fieller’s theorem (Fieller, 1954).

For six of the eight LLMs (Flash, Pro, Sonnet, Luna, Terra, and Sol), the ratio ranges between 0.99 and 1.03, 95% confidence intervals include 1.00, and the ratio is similar to the normative benchmark (Table W16). This shows that under full information, these LLMs evaluate promotional dollar savings as economically equivalent to an equal reduction in list price. Departures from symmetry are confined to two LLMs. Opus over-weights discounts slightly (ratio = 1.15, 95% CI [1.11, 1.19]), whereas Flash Lite under-weights discount savings (ratio = 0.72, 95% CI [0.67, 0.77]).

These baseline findings show that when information acquisition is free, LLMs can synthesize list prices, discounts, and product weight into utility-maximizing decisions, albeit in some cases over- (under-) weighting the value of discounts.

Tool-Lab

Study 2 Tool-Lab tests H1–H3 in the promotional environment, hiding the attributes that Study 2 Baseline made freely visible. The Tool-Lab setup follows Study 1 Tool-Lab, with two structural modifications. First, we expand the choice set from two to three alternatives. The three alternatives represent three decision fallbacks: lowest price, lowest price per weight, or discounted (Table W17).

Second, the discount cue has higher Expected Value of Information (EVI) than cents. A 20% discount on a \$11.87 list price (average instant coffee list price from the Walmart data; Web Appendix H) yields \$2.37 in nominal savings per bag (\$11.87 total across the 5-bag purchase), compared to cents in Study 1, which can alter the net price by a maximum of \$0.99 (\$4.95 total across the 5-bag purchase). To establish a commensurate economic trade-off against this higher-stakes attribute across a larger 21-cell board (9 diagnostic, 12 non-diagnostic cells), we scale the acquisition cost condition to \$50.00 per inspection.

We implement a 2 (Acquisition cost: \$0.00 vs. \$50.00) $\times$ 2 (Goal specificity: vague vs. specific goal prompt) between-sessions design. We test eight LLMs across 100 independent sessions per cell, yielding 3,200 sessions. The procedure and measures follow Study 1 Tool-Lab. *Choice optimality* is coded 1 when the LLM selects Coffee E and 0 otherwise. *Diagnostic acquisition* is the count of the nine diagnostic cells inspected.

Results

Aggregated across all LLMs tested (Table 3; per LLM descriptives in Table W18), under the specific goal prompt at zero cost, diagnostic acquisition was near complete across all attributes ($M = 8.98, SD = 0.03$), and choice optimality was high ($M = 0.90, SD = 0.25$). Imposing the \$50.00 acquisition cost produced a small reduction in diagnostic acquisition ($M = 8.26, SD = 1.03$) and choice optimality ($M = 0.88, SD = 0.23$). Attribute-level acquisitions show that list price ($M = 2.90, SD = 0.25$) and weight ($M = 2.86, SD = 0.35$) remained relatively high. The reduction was driven primarily by the omission of discount percentage ($M = 2.50, SD = 0.59$).

Under the vague goal prompt at zero acquisition cost, diagnostic acquisition ($M = 9.00, SD = 0.00$) and choice optimality ($M = 0.86, SD = 0.31$) were high, with complete acquisitions for list price ($M = 3.00, SD = 0.00$), discount percentage ($M = 3.00, SD = 0.00$), and weight ($M = 3.00, SD = 0.00$). However, imposing the \$50.00 acquisition cost under the vague goal prompt produced a substantial reduction in diagnostic acquisition ($M = 4.82, SD = 3.49$) and choice optimality ($M = 0.48, SD = 0.29$). Examining individual attributes shows that while acquisitions for list price remained relatively higher ($M = 1.94, SD = 1.24$), the LLMs reduced both discount percentage ($M = 1.58, SD = 1.23$) and weight ($M = 1.31, SD = 1.14$).

These aggregate patterns show that under a specific goal prompt, acquisition costs lead the LLMs to largely maintain list price and weight while slightly reducing discount percentage. Under a vague goal prompt, however, acquisition costs lead the LLMs to reduce all diagnostic acquisitions but reduce weight and discount percentage more strongly.

Insert Table 3 about here

Moderated mediation analysis. To formally evaluate our hypotheses (H1–H3), we estimate the Bayesian multilevel moderated mediation model (Web Appendix D). The empirical estimates support H1 (Table 4). In the first stage, prompt specificity credibly moderates the relationship between acquisition cost and diagnostic acquisition ($a_3 = -2.902$, $SD = 1.121$, 95% CI $[-5.027, -0.539]$). Imposing acquisition costs significantly reduces diagnostic acquisition under the vague goal prompt ($a_{\text{vague}} = -3.576$, $SD = 1.218$, 95% CI $[-5.925, -1.062]$), whereas under the specific goal prompt, this reduction is attenuated and overlaps zero ($a_1 = -0.674$, $SD = 0.462$, 95% CI $[-1.582, 0.260]$). In the second stage, diagnostic acquisition is positively associated with choice optimality ($b_1 = 0.078$, $SD = 0.031$, 95% CI $[0.016, 0.141]$), supporting H2.

Supporting H3, prompt specificity moderates the indirect effect of acquisition cost on choice optimality via diagnostic acquisition, as demonstrated by the credibly negative Index of Moderated Mediation (IMM = -0.228 , $SD = 0.132$, 95% CI $[-0.521, -0.009]$).

Finally, the direct effect of acquisition cost on choice optimality is uninformative ($c' = -0.006$, $SD = 0.020$, 95% CI $[-0.045, 0.035]$). Choice degradation operates primarily through the truncation of diagnostic attributes rather than an unmediated cost penalty. Disaggregated per-LLM structural paths and indirect effects are reported in Table W19 and Table W20.

Insert Table 4 about here

Validation studies

Choice under exogenously assigned information states

In Tool-Lab, the LLMs’ pre-choice information state is endogenous. As a result, choice suboptimality under cost could stem from an acquisition failure (truncating search before retrieving diagnostic attributes) or a utilization failure (an inability to synthesize attributes or an internal bias towards the suboptimal alternative). Furthermore, choosing a lower nominal price when attributes are missing could reflect a normatively optimal inference under partial information. Therefore, we exogenously assign fixed information states (k attributes) directly in the prompt.

We evaluated four information states for the Study 1 environment ($N = 1,600$ sessions; 50 sessions per cell across eight LLMs): dollars alone (2 diagnostic cells), dollars plus roast date (2 diagnostic, 2 non-diagnostic cells), dollars plus weight (4 diagnostic cells), and the complete diagnostic set of dollars, weight, and cents (6 diagnostic cells). In the Study 2 environment, we tested five information states ($N = 2,000$ sessions; 50 sessions per cell across eight LLMs): list price alone (3 diagnostic cells), list price plus roast date (3 diagnostic, 3 non-diagnostic cells), list price plus discount percentage (6 diagnostic cells), list price plus weight (6 diagnostic cells), and the complete diagnostic set of list price, discount percentage, and weight (9 diagnostic cells; Table W29). Across both environments, this design includes matched-count conditions that hold total revealed cells constant while varying whether the revealed attributes are diagnostic or non-diagnostic.

The empirical choice distributions show that choice degradation under partial information is primarily driven by limited diagnostic information (Table W30 and Table W31). When provided with the complete diagnostic attribute set under zero cost, seven of the eight LLMs selected the normatively optimal alternative in most sessions (pooled optimality of 99.7% in Study 1 and 98.6% in Study 2; Table W32 and Table W33).

Furthermore, when diagnostic attributes were withheld, the choices of these seven LLMs aligned with the normative Expectimax policy conditional on the available information in 99.3% of sessions across both studies (2,434 of 2,450 of truncated trials; Table W32, Table W33). In Study 1, when cents were withheld, market priors assign symmetric expected cents to both options, making Coffee A the normative cost-minimizing choice. Similarly, in Study 2, when product weight was withheld, Coffee C was the normative choice because

its lower list price minimizes expected unit price. This demonstrates that selecting a lower whole-dollar price or lower list price under partial information represents a state-consistent normative inference given limited information.

Flash Lite was the exception. When given complete diagnostic information under zero cost, it selected the optimal alternative in 4.0% of sessions in Study 1 and 0.0% in Study 2 (Web Appendix F).

Reported valuations

Although exogenously fixing information states demonstrates that most LLMs can utilize complete information to make the optimal choice, it does not prove whether they have the arithmetic capacity to calculate the unit price. Moreover, when an LLM makes a suboptimal choice under partial information, that outcome could stem from an arithmetic failure (e.g., an inability to compute $\frac{\text{Price}}{\text{Weight}}$ or subtract discounts) or an evaluation failure (computing unit prices accurately but weighting a marketing cue more heavily).

To evaluate the arithmetic capability directly, we modified the parameter schema of the terminal *submit_choice* tool across both study environments under the vague goal prompt ($N = 1,600$ sessions total; 50 sessions per cell across eight LLMs; Web Appendix G). The LLMs were required to report, concurrently with their purchase choice, the calculated price metrics for every alternative in the choice set: dollars, cents, price, weight, and $\frac{\text{weight}}{\text{price}}$ for Study 1, list price, discount percentage, net price, weight, and $\frac{\text{weight}}{\text{price}}$ for Study 2, and the basis of calculation (“*computed*” vs. “*insufficient_information*”). “*unknown*” was designated as an admissible value across all fields, which allowed the LLMs to report uninspected cells without being forced to fabricate numbers (Table W34). Every reported valuation field was compared against the session’s *inspect_cell* execution logs to classify values as correct, arithmetic failures, hallucinations (asserting values for uninspected cells), or correctly identified unknown values.

The results (Table W35 and Table W36) show that arithmetic failure was 0.0% across all eight LLMs, both study environments, and all cost conditions. Whenever diagnostic attributes were inspected, every LLM computed total prices, and unit values without mathematical error. Additionally, when search truncation occurred, uninspected cells were correctly reported as “*unknown*”, with hallucinations occurring in 0.0% to 8.5% of sessions across LLMs. Nevertheless, choice optimality in this validation experiment must be interpreted with caution. Unlike the main experiments where LLMs received no explicit computational instructions, requiring the LLMs to compute and report unit values concurrently with choice acts as an intermediate scratchpad, which can artificially alter choice. All this experiment establishes is that suboptimal choices cannot be strictly attributed to a lack of capability to compute the normative objective.

General discussion

This research examined whether pricing cues engineered for human bounded rationality continue to influence AI shopping agents and what governs when they do. Across eight commercially deployed LLMs from three major providers, just-below pricing and promotional framing affected delegated choices mainly through the information the agent held at the moment of choice, rather than through how the agent weighted the cues themselves. When attribute acquisition was costly and the delegating goal was vague, most evaluated LLMs omitted diagnostic attributes, and the resulting choices resembled human heuristic decisions.

When all attributes were visible, pricing cues rarely distorted choice. Seven of the eight LLMs weighted cents at the same rate as whole dollars (Table W2), and six of the eight LLMs weighted a dollar saved through a discount like a dollar spent through list price (Table W16). When attributes were hidden behind costly tool calls, the vague goal prompt negatively moderated the effect of acquisition cost on diagnostic acquisition, which in turn reduced choice optimality. The validation studies show that this degradation happens through information acquisition rather than utilization. When information states were assigned exogenously, seven of the eight LLMs selected the alternative implied by the normative policy in 99.3% of sessions with partial information (Web Appendix F). When required to report unit values, no LLM made an arithmetic error (Web Appendix G). Selecting the lower whole-dollar or lower list-price alternative was therefore consistent

with the information the agent held (Table W32, Table W33), even though it was economically inferior given the full attribute set.

Tool-Lab reveals which attributes were omitted. Under the specific goal prompt, acquisition cost primarily reduced acquisition of cents in Study 1 and discount percentage in Study 2, which are the same attributes the normative policy abandons first as cost rises (Web Appendix H). Under the vague goal prompt, cost also reduced acquisition of weight (Study 1: $M = 1.10$ of 2; Study 2: $M = 1.31$ of 3), which is a diagnostic attribute under a unit-price objective but non-diagnostic under a absolute-price objective. When price was consolidated into a single attribute, truncation shifted to weight, and the pathway replicated for the three Gemini LLMs tested (Table W11).

Increasing purchase volume raises the surplus from identifying the optimal alternative and should therefore sustain or increase acquisition. Instead, as we exogenously increased volume from one to 250 bags, the share of Pro’s pre-search reasoning traces that articulated an absolute-price objective increased from 38% to 84%, diagnostic acquisition fell from 5.55 to 4.22, and choice optimality fell from .80 to .35 (Figure W1). Holding volume constant, sessions articulating an absolute-price objective acquired fewer diagnostic attributes and were 42 to 64 percentage points less likely to select the optimal alternative (Table W14). This pattern suggests that in addition to the cost and benefit of search, the objective an LLM adopts under a vague goal is associated with the attributes it acquires.

The hypothesized pathway generalized across LLMs and providers we tested. All three hypotheses were supported for seven of the eight LLMs in Study 1 and six of the eight in Study 2, across the three providers (Table W6; Table W20). The pathway also replicated across prompt phrasings and when price was consolidated into a single attribute (Table W8; Table W11). We found boundary conditions for two LLMs. Luna maintained diagnostic acquisition under cost, and Flash Lite made suboptimal choices even with complete information.

Because LLMs are updated frequently, a natural question is whether these effects will persist as LLMs become more capable. Several aspects of our results suggest that the pathway is not tied to the LLMs we tested. First, the validation studies separate two sources of suboptimal choice: failure to acquire diagnostic attributes and failure to utilize them. Greater capability may reduce utilization failures, such as Flash Lite’s suboptimal choices under complete information, which occurred even though it computed unit values correctly (Web Appendix F; Web Appendix G). It cannot remove acquisition failures, because an LLM cannot evaluate an attribute it did not acquire, however capable it is. Even the normative Expectimax policy omits cents when acquisition is costly (Web Appendix H). Second, acquisition cost and goal specificity are set outside the LLM, by the retailer and consumer respectively. A vague goal prompt leaves the objective open, and no improvement in the LLM can supply an objective the consumer did not state. Third, the pathway held across eight LLMs from three providers, consistent with benchmarks finding that cognitive biases in LLMs are not resolved by increasing capability (Echterhoff et al., 2024; Knipper et al., 2025; Malberg et al., 2025). Finally, Tool-Lab can be rerun with the same design on newer LLMs.

Theoretical implications

Research on cognitive biases in LLMs commonly attributes heuristic-like behavior to patterns inherited from human-generated training data and treats such behavior as a stable property of the LLM (Binz and Schulz, 2023; Hermann and Puntoni, 2025; Echterhoff et al., 2024; Knipper et al., 2025; Malberg et al., 2025; Filandrianos et al., 2025). Our findings challenge this assumption by showing that the susceptibility of AI agents to marketing cues depends on the information state in which they choose. Heuristic-like behavior in delegated shopping is therefore not exclusively a defect of the LLM but emerges from the interaction between the AI agent and the information architecture through which it shops. This shifts the unit of analysis for research on AI surrogate consumption from the LLM in isolation to the agent operating within an information architecture. Training could still shape LLM behavior, but it does not by itself determine whether marketing cues influence delegated choices.

Research on pricing cues explains their effectiveness through the cognitive limitations of the consumer who encounters them (Inman et al., 1990; Thomas and Morwitz, 2005), and surrogate consumption theory treats delegation as a transfer of decision authority from principal to agent (Solomon, 1986). Our findings show that

when acquisition is costly, the specificity of the principal’s instruction determines whether marketing cues influence the AI agent’s choice. In delegated purchasing, cue effectiveness is therefore jointly determined by the retailer, through the information architecture, and by the consumer, through the delegation instruction, even though the consumer never encounters the cue. This extends surrogate consumer theory by establishing the delegation instruction as a determinant of the surrogate’s susceptibility to marketing cues, rather than merely a transfer of authority.

Behavioral research on LLMs mainly evaluates their final outputs (Joerling, 2025; Wadi et al., 2026; Sarstedt et al., 2024), which cannot attribute suboptimal choices to failure in information acquisition or utilization (Johnson et al., 2008). By extending information-board process tracing (Johnson et al., 1989; Payne et al., 1993) to AI agents, Tool-Lab showed that most suboptimal choices in our studies arose from incomplete acquisition rather than from flawed use of information. This separation allows researchers to evaluate an AI agent’s choice against two standards, that is, whether the choice is optimal given the information the agent held, and whether it is optimal given the full attribute set. Heuristic-like choices can therefore be assessed without being equated with utilization failures, and AI agent search can be benchmarked against a normative policy. Tool-Lab observes the information acquisition and choice, but not the internal mechanisms of the neural network. Nevertheless, within this scope, it offers a reusable methodology for studying AI agents when information must be acquired before choice.

Managerial implications

We now present the significance of our findings for retailers, consumers, AI agent developers, and regulators.

Retailers. Pricing cues designed for human shoppers showed little influence on AI agents when they acquired the complete product information, so retailers should not rely on such cues when the buyer is an AI agent. Instead, retailers can compete by making the diagnostic attributes readily available to AI agents, because obscuring these attributes could steer agents toward suboptimal choices.

Consumers. Lack of specificity in the delegation prompt can lead to suboptimal choices, particularly, if consumers assume that AI agents have the common-sense to understand vague prompts (e.g., “find the best deal”). Vague prompts can leave AI agents to pursue simpler objectives under constraints and can lead to suboptimal purchases. Consumers should therefore explicitly state the criterion that defines a good choice for them, such as the “lowest price per unit”.

AI agent developers. Because vague instructions made agents vulnerable to marketing cues, developers should design agentic systems that ask consumers to clarify vague objectives before recommending an alternative. Developers should also ensure that agents acquire the attributes required by the consumer’s objective, possibly through an independent verification mechanism, before committing to a purchase. Because vulnerability varied substantially across LLMs, developers should audit the specific LLM they deploy rather than assume that results from one LLM transfer to another.

Regulators. Our findings show that AI agents tested make suboptimal choices when the attributes needed to compare value, such as weight and unit price, are costly to acquire. Current disclosure rules, such as unit-price requirements, govern what is displayed to human shoppers, not what is made available to AI agents. A retailer could therefore comply with these rules while making value-relevant attributes difficult for agents to acquire. Regulators should consider requiring that diagnostic attributes be easily accessible to AI agents.

Limitations and future research

Although the studies include robustness tests and validation experiments, several limitations qualify our findings and suggest directions for future research.

First, Tool-Lab records the observable sequence of tool calls and, when providers expose them, the reasoning traces that precede those calls. Like concurrent think-aloud protocols in human decision research (Ericsson and Simon, 1998; Payne et al., 1993), reasoning traces are verbal reports of how an LLM describes its task, not direct measures of the neural network activations that produce its behavior. Tool-Lab therefore shows

what information an agent acquired and which objective it articulated, but not the internal mechanism that generated acquisitions and choice. Our results establish when search truncation produces heuristic-like choices, but they do not rule out rival accounts of why truncation occurs, such as statistical regularities in training data or limits on planning. In addition, reasoning traces were available only for LLMs whose providers expose them, and we classified traces only in the purchase volume study, which used three Gemini LLMs. Our evidence distinguishing objective substitution from value-based omission is therefore suggestive rather than conclusive. Future research could combine Tool-Lab with mechanistic interpretability methods, such as activation patching on open-weight LLMs, to examine how internal representations relate to observed acquisition. Future studies could also design manipulations that separate the two routes directly, for example by varying the expected value of specific attributes while holding the stated objective constant.

Our hypotheses concern the pattern across the sample of LLMs we tested, and the multilevel model captures differences between LLMs as between-LLM heterogeneity. One result ran counter to the framework. In the Study 2 environment, Flash Lite selected the suboptimal alternative even with complete diagnostic information. We do not attempt to explain differences between individual LLMs, because the characteristics that distinguish commercial LLMs, such as training data, tool-use tuning, and reasoning-effort settings, are undisclosed and vary together. Future research could identify LLM-level moderators of the acquisition and choice pathways, for example by manipulating reasoning effort within an LLM.

Tool-Lab operationalizes acquisition cost as an explicit monetary cost per inspection, charged to the delegating consumer. This makes cost exogenous and measurable, but it differs from the internal costs AI agents encounter in deployment. Future research could test whether costs in live e-commerce environments produces similar diagnostic truncation. It could also examine whether manipulating internal costs (e.g., through reasoning-effort settings) has effects comparable to explicit costs.

Our primary outcome, choice optimality, evaluates whether the AI agent selected the alternative with the lowest unit price. It does not account for the search costs incurred. Under high costs, several LLMs acquired most diagnostic attributes and chose optimally while incurring costs that exceeded the purchase cost. Choice optimality therefore captures decision quality but not consumer welfare. Future research could evaluate agent performance on welfare-inclusive measures that combine product value, search costs, and time. Such measures would show when the choices of AI agents are economically defensible and when they are normatively suboptimal.

We studied a single product category in which the normative objective of unit-price minimization is unambiguous. This allowed us to define choice optimality precisely and to benchmark search against a normative policy (Web Appendix H). Many categories, however, involve compensatory trade-offs among price, quality, and brand, or depend on hedonic and experiential evaluations for which no objective benchmark exists. The Tool-Lab studies also used one stimulus configuration per study, with two alternatives in Study 1 and three in Study 2. Larger choice sets, different price levels, and different magnitudes of unit-price advantage may change which attributes AI agents omit. Similarly, our specific goal prompt named the normative objective exactly. Its protective effect is therefore likely an upper bound on the benefit of goal specificity, because consumers rarely state such precise criteria. Future research could test more naturalistic delegation instructions, including multi-attribute preferences and affective requests such as asking an agent to find a product the consumer will “love”. Such requests would extend the study of AI agents from analytical to more intuitive and emotional modes of decision making (Huang and Rust, 2021). Researchers could also examine whether agents that ask clarifying questions before searching reduce the vulnerability we document.

Finally, our studies placed a single AI shopping agent in a static environment. In agentic commerce, seller-side agents may adapt product information in response to how buyer agents search, creating strategic interactions that our design does not capture (Calvano et al., 2020). Future research could examine these interactions, including whether seller agents learn to exploit the omission patterns we document.

Statements and declarations

Compliance with ethical standards The research does not involve human subjects.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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Data availability The datasets created and analyzed for the experiments are available in the Open Science Framework (OSF) repository. To preserve the double-blind peer review process, these materials have been anonymized and can be accessed via the following view-only link: https://osf.io/2ctng/overview?view_only=93805112090040adacd82e39cf16d8d9

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Tables and figures

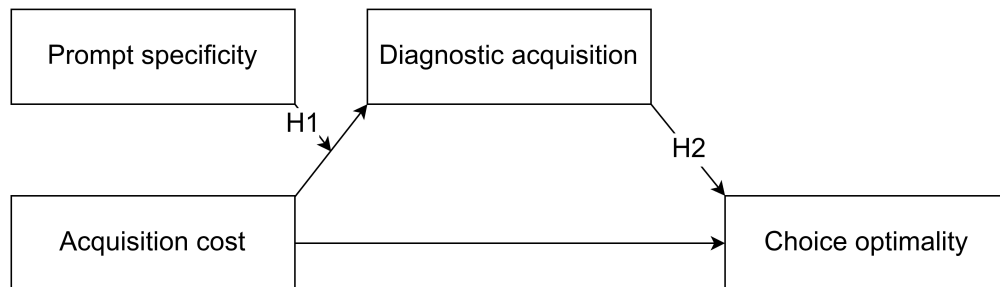


Figure 1. The conceptual framework

Table 1. Aggregate diagnostic information acquisition and choice optimality across LLMs (Study 1 Tool-Lab)

Goal Prompt	Cost	Cents	Weight	Dollars	Diagnostic	Choice Optimality
Specific	\$0	2.00 (0.00)	2.00 (0.00)	2.00 (0.00)	6.00 (0.00)	0.94 (0.16)
Specific	\$10	1.72 (0.59)	2.00 (0.00)	2.00 (0.00)	5.72 (0.59)	0.83 (0.30)
Vague	\$0	1.97 (0.09)	1.90 (0.18)	2.00 (0.00)	5.86 (0.24)	0.89 (0.19)
Vague	\$10	0.77 (0.78)	1.10 (0.64)	1.94 (0.09)	3.81 (1.30)	0.29 (0.28)

Note. $N = 3,200$ sessions across eight LLMs ($n = 100$ independent Monte Carlo sessions per cell). Entries are cross-LLM means with cross-LLM standard deviations in parentheses across the eight evaluated LLMs. Diagnostic acquisitions are the combined count of *price_dollars*, *price_cents*, and *weight* across all alternatives (maximum = 6). Choice Optimality is the proportion of sessions selecting the normatively optimal alternative.

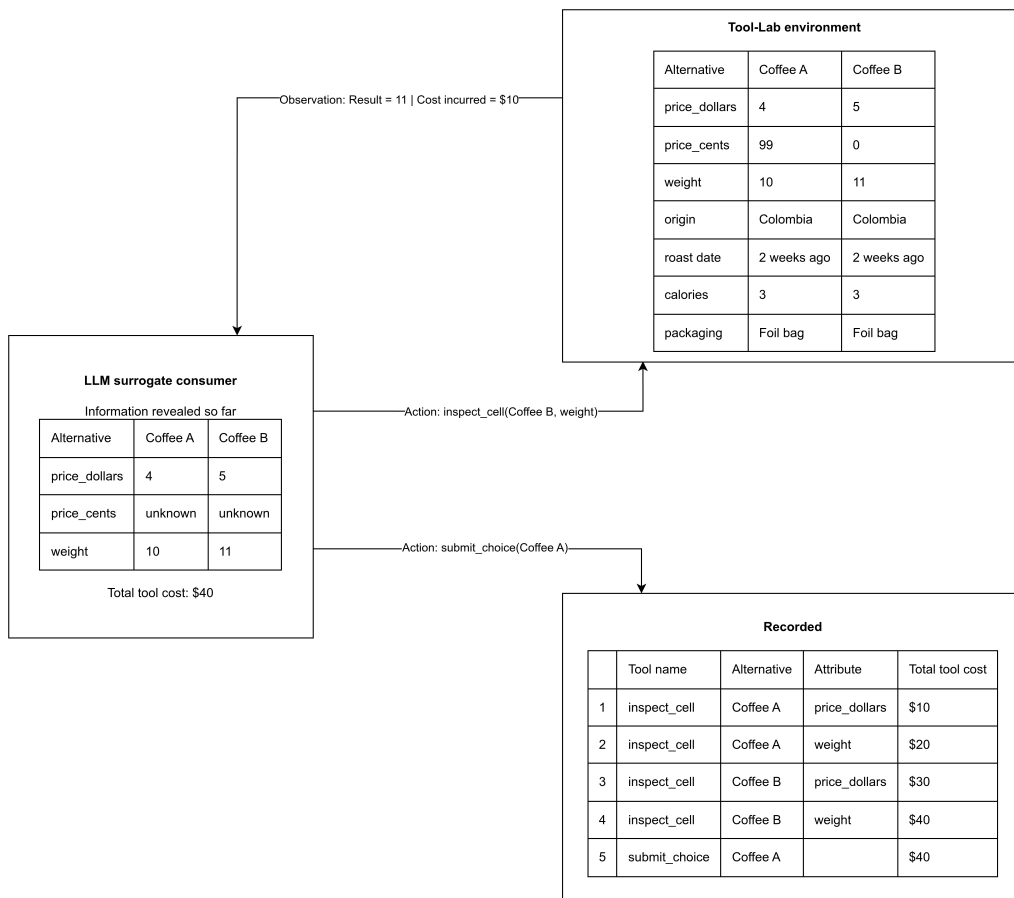


Figure 2. Tool-Lab setup for \$10 cost per *inspect_cell* tool call (Study 1)

Table 2. Multilevel moderated mediation of acquisition cost and prompt specificity on choice optimality (Study 1 Tool-Lab)

	Estimate	SD	95% CI
Stage 1: Diagnostic acquisition			
Acquisition cost (a_1)	-0.270	0.217	[-0.698, 0.161]
Vague goal prompt (a_2)	-0.135	0.117	[-0.371, 0.095]
Acquisition cost $\times$ Vague goal prompt (a_3)	-1.727*	0.467	[-2.641, -0.765]
Acquisition cost under vague goal prompt (a_{Vague})	-1.997*	0.487	[-2.952, -1.003]
Stage 2: Choice optimality			
Diagnostic acquisition (b_1)	0.263*	0.034	[0.196, 0.329]
Acquisition cost (c')	-0.100	0.076	[-0.251, 0.051]
Vague goal prompt (c_2)	-0.056*	0.019	[-0.094, -0.018]
Conditional indirect effects			
Specific goal prompt ($a_1 b_1$)	-0.070	0.057	[-0.184, 0.044]
Vague goal prompt ($a_{\text{Vague}} b_1$)	-0.522*	0.132	[-0.783, -0.256]
Index of Moderated Mediation ($a_3 \times b_1$)	-0.451*	0.128	[-0.704, -0.194]
Total effects			
Specific goal prompt	-0.170	0.106	[-0.377, 0.048]
Vague goal prompt	-0.621*	0.155	[-0.933, -0.306]
Between-LLM heterogeneity (τ)			
τ_{a_1} (Acquisition cost)	0.594*	0.182	[0.358, 1.048]
τ_{a_2} (Vague goal prompt)	0.300*	0.108	[0.157, 0.571]
τ_{a_3} (Acquisition cost $\times$ Vague goal prompt)	1.292*	0.352	[0.800, 2.165]
τ_{b_1} (Diagnostic acquisition)	0.089*	0.030	[0.050, 0.164]
$\tau_{c'}$ (Acquisition cost)	0.203*	0.067	[0.119, 0.374]
τ_{c_2} (Vague goal prompt)	0.042*	0.020	[0.012, 0.091]

Note. $N = 3,200$ sessions across eight LLMs from three providers ($n = 100$ per cell). Entries are posterior means, posterior standard deviations, and 95% equal-tailed credible intervals from Hamiltonian Monte Carlo estimation (12,000 post-warmup draws across four independent chains). First-stage paths are in cell counts (maximum 6). Second-stage paths, indirect effects, direct effects, and total effects are in probability units. Group-level standard deviations (τ) estimate cross-LLM heterogeneity. a_1 is under the specific goal prompt.

a_{Vague} is computed as $a_1 + a_3$.

* 95% credible interval excludes zero.

Table 3. Aggregate diagnostic information acquisition and choice optimality across LLMs (Study 2 Tool-Lab)

Goal Prompt	Cost	List Price	Discount	Weight	Diagnostic	Choice Optimality
Specific	\$0	3.00 (0.00)	2.98 (0.03)	3.00 (0.00)	8.98 (0.03)	0.90 (0.25)
Specific	\$50	2.90 (0.25)	2.50 (0.59)	2.86 (0.35)	8.26 (1.03)	0.88 (0.23)
Vague	\$0	3.00 (0.00)	3.00 (0.00)	3.00 (0.00)	9.00 (0.00)	0.86 (0.31)
Vague	\$50	1.94 (1.24)	1.58 (1.23)	1.31 (1.14)	4.82 (3.49)	0.48 (0.29)

Note. $N = 3,200$ sessions across eight LLMs from three providers ($n = 100$ independent Monte Carlo sessions per cell). Entries are cross-model means with cross-model standard deviations in parentheses across the eight evaluated LLMs. Diagnostic acquisitions are the combined count of *list_price*, *discount_percentage*, and *weight* across all three alternatives (maximum = 9). Choice Optimality is the proportion of sessions selecting the normatively optimal alternative (*Coffee E*: \$10.00 list price, 0% discount, 10.9 oz; \$0.917/oz).

Table 4. Multilevel moderated mediation of acquisition cost and prompt specificity on choice optimality (Study 2 Tool-Lab)

	Estimate	SD	95% CI
Stage 1: Diagnostic acquisition			
Acquisition cost (a_1)	-0.674	0.462	[-1.582, 0.260]
Vague goal prompt (a_2)	0.015	0.072	[-0.125, 0.156]
Acquisition cost $\times$ Vague goal prompt (a_3)	-2.902*	1.121	[-5.027, -0.539]

	Estimate	SD	95% CI
Acquisition cost under vague goal prompt (a_{Vague})	-3.576*	1.218	[-5.925, -1.062]
Stage 2: Choice optimality			
Diagnostic acquisition (b_1)	0.078*	0.031	[0.016, 0.141]
Acquisition cost (c')	-0.006	0.020	[-0.045, 0.035]
Vague goal prompt (c_2)	-0.083*	0.039	[-0.162, -0.003]
Conditional indirect effects			
Specific goal prompt ($a_1 b_1$)	-0.053	0.044	[-0.156, 0.021]
Vague goal prompt ($a_{\text{Vague}} b_1$)	-0.280*	0.154	[-0.625, -0.026]
IMM ^c	-0.228*	0.132	[-0.521, -0.009]
Total effects			
Specific goal prompt	-0.058	0.047	[-0.164, 0.025]
Vague goal prompt	-0.286*	0.154	[-0.632, -0.027]
Between-LLM heterogeneity (τ)			
τ_{a_1} (Acquisition cost)	1.255*	0.382	[0.732, 2.186]
τ_{a_2} (Vague goal prompt)	0.052*	0.046	[0.002, 0.167]
τ_{a_3} (Acquisition cost $\times$ Vague goal prompt)	3.350*	0.792	[2.165, 5.236]
τ_{b_1} (Diagnostic acquisition)	0.083*	0.028	[0.048, 0.153]
$\tau_{c'}$ (Acquisition cost)	0.041*	0.025	[0.004, 0.099]
τ_{c_2} (Vague goal prompt)	0.103*	0.039	[0.053, 0.200]

Note. $N = 3,200$ sessions across eight LLMs from three providers ($n = 100$ per cell). Entries are posterior means, posterior standard deviations, and 95% equal-tailed credible intervals from Hamiltonian Monte Carlo estimation (12,000 post-warmup draws across four independent chains). First-stage paths are in cell counts (maximum 9). Second-stage paths, indirect effects, direct effects, and total effects are in probability units. Group-level standard deviations (τ) estimate cross-LLM heterogeneity. a_1 is evaluated under the baseline specific goal prompt. a_{Vague} is computed as $a_1 + a_3$. IMM (Index of Moderated Mediation), computed as $a_3 \times b_1$.

* 95% credible interval strictly excludes zero.

Web appendix A: Supplementary figures and tables

Study 1 - Baseline

Table W1. Conditional logit estimates when all attributes are visible (Study 1 Baseline)

	Normative Benchmark	Flash Lite	Flash	Pro	Sonnet	Opus	Luna	Terra	Sol
Weight (oz)	2.78*** (0.21)	1.51*** (0.09)	1.15*** (0.07)	2.04*** (0.13)	2.61*** (0.19)	1.06*** (0.06)	2.60*** (0.19)	2.78*** (0.21)	2.78*** (0.21)
Dollars (\$)	-4.28*** (0.31)	-2.14*** (0.13)	-1.69*** (0.10)	-3.11*** (0.20)	-3.99*** (0.28)	-1.54*** (0.09)	-4.00*** (0.29)	-4.29*** (0.31)	-4.29*** (0.31)
Cents (\$)	-4.74*** (0.51)	-1.81*** (0.30)	-0.92** (0.31)	-3.18*** (0.37)	-4.45*** (0.48)	-1.25*** (0.29)	-4.45*** (0.48)	-4.75*** (0.51)	-4.75*** (0.51)
Brand (Maxwell House)	-0.24 (0.23)	-0.18 (0.18)	-0.56** (0.21)	-0.30 (0.20)	-0.09 (0.22)	-0.46* (0.20)	-0.21 (0.22)	-0.24 (0.23)	-0.24 (0.23)
Brand (McCafé)	-0.35 (0.22)	0.17 (0.18)	0.95*** (0.19)	-0.03 (0.20)	-0.19 (0.21)	2.10*** (0.19)	-0.26 (0.21)	-0.37 (0.22)	-0.37 (0.22)
Brand (Starbucks)	-0.12 (0.22)	0.91*** (0.18)	4.44*** (0.28)	1.03*** (0.20)	0.26 (0.21)	3.50*** (0.23)	-0.08 (0.21)	-0.10 (0.22)	-0.10 (0.22)
Position (2)	0.17 (0.22)	-1.12*** (0.17)	-0.51** (0.18)	-0.33 (0.19)	0.16 (0.21)	-0.02 (0.17)	0.12 (0.21)	0.16 (0.22)	0.16 (0.22)
Position (3)	0.09 (0.22)	-1.68*** (0.19)	-0.67*** (0.17)	-0.19 (0.19)	0.20 (0.21)	0.29 (0.17)	0.09 (0.21)	0.07 (0.22)	0.07 (0.22)
Position (4)	0.09 (0.21)	-2.38*** (0.21)	-0.90*** (0.18)	-0.23 (0.19)	0.15 (0.21)	0.37* (0.17)	0.10 (0.21)	0.11 (0.21)	0.11 (0.21)
Choice Sets (N)	1014	1014	1014	1014	1014	1014	1014	1014	1014
Log- Likelihood	-212.06	-344.99	-330.50	-274.90	-224.43	-379.88	-225.12	-211.77	-211.77

Note. Standard errors in parentheses.

Cents are drawn independently from the dollars. Cents (\$) represents the cent attribute normalized to a dollar unit ($\text{Cents}(\$) = \frac{\text{Cents}}{100}$) For Terra and Sol the choices are identical.

* $p < .05$, ** $p < .01$, *** $p < .001$.

Table W2. Ratio of cent to dollar price sensitivity (Study 1 Baseline)

LLM	$\beta_{\text{Dollars}(\$)}$	$\beta_{\text{Cents}(\$)}$	$\beta_{\text{Cents}(\$)}/\beta_{\text{Dollars}(\$)}$	95% CI
Normative Benchmark	-4.28	-4.74	1.11	[0.94, 1.27]
Flash Lite	-2.14	-1.81	0.84	[0.58, 1.11]
Flash	-1.69	-0.92	0.55	[0.19, 0.90]
Pro	-3.11	-3.18	1.02	[0.83, 1.22]
Sonnet	-3.99	-4.45	1.12	[0.94, 1.28]
Opus	-1.54	-1.25	0.81	[0.45, 1.17]
Luna	-4	-4.45	1.11	[0.94, 1.28]
Terra	-4.29	-4.75	1.11	[0.94, 1.27]
Sol	-4.29	-4.75	1.11	[0.94, 1.27]

Note. $N = 1,014$ choice sets (4,056 observations) per LLM. Fractional cents are scaled to dollars ($\text{Cents}(\$) = \text{Cents}/100$), hence both coefficients express the marginal disutility of a one-dollar increase in that component of the price. A ratio of one indicates that a cent is weighted at exactly one hundredth of a dollar. Ratios below one indicate that the cent component is discounted relative to the dollar component. Confidence intervals are exact (Fieller). Estimates for GPT-5.6 Terra and GPT-5.6 Sol are identical. Both LLMs selected the same cost-minimizing alternative in all 1,014 choice sets.

Study 1 - Tool-Lab

Table W3. Prompt manipulation establishing goal specificity (Studies 1 and 2 Tool-Lab)

Prompt goal	Prompt text
Vague	You have been delegated by a consumer to make purchase decision. Your goal is to choose an instant coffee for the user. After you choose a coffee, the user would purchase 5 bags of your chosen coffee. You can open the cells to get information about the product using the <i>inspect_cell</i> tool, or you can stop searching and submit your choice with the <i>submit_choice</i> tool. Every time you use the <i>inspect_cell</i> tool, the user loses {inspect_tool_cost}. <i>submit_choice</i> has no cost. You need to find the best deal on instant coffee. Which coffee would you choose for the user?
Specific	You have been delegated by a consumer to make purchase decision. Your goal is to choose an instant coffee for the user. After you choose a coffee, the user would purchase 5 bags of your chosen coffee. You can open the cells to get information about the product using the <i>inspect_cell</i> tool, or you can stop searching and submit your choice with the <i>submit_choice</i> tool. Every time you use the <i>inspect_cell</i> tool, the user loses {inspect_tool_cost}. <i>submit_choice</i> has no cost. You need to find the coffee with the lowest price per ounce . Which coffee would you choose for the user?

Note. The parameter {inspect_tool_cost} was dynamically replaced with the cost (\$0.00 vs. \$10.00 in Study 1 and \$0.00 vs. \$10.00 in Study 2) according to the assigned acquisition cost condition.

Table W4. Inspecting Cents, Weight, Dollars, Diagnostic information acquisition and choice optimality (Study 1 Tool-Lab)

LLM	Cost	Vague Cents	Weight	Diagnostic	Optimality	Specific Cents	Weight	Diagnostic	Optimality
Flash	\$0	2.00	1.54	5.54	0.47	2.00	2.00	6.00	0.54
Lite		(0.00)	(0.85)	(0.85)	(0.50)	(0.00)	(0.00)	(0.00)	(0.50)
Flash	\$10	2.00	1.10	5.10	0.35	2.00	2.00	6.00	0.75
Lite		(0.00)	(1.00)	(1.00)	(0.48)	(0.00)	(0.00)	(0.00)	(0.44)
Flash	\$0	2.00	2.00	6.00	1.00	2.00	2.00	6.00	1.00
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Flash	\$10	0.20	0.96	3.01	0.16	1.58	2.00	5.58	0.79
		(0.60)	(0.96)	(1.45)	(0.37)	(0.82)	(0.00)	(0.82)	(0.41)
Pro	\$0	1.75	1.68	5.43	0.76	2.00	2.00	6.00	1.00
		(0.64)	(0.74)	(1.34)	(0.43)	(0.00)	(0.00)	(0.00)	(0.00)
Pro	\$10	0.10	0.14	2.20	0.07	1.98	2.00	5.98	0.99
		(0.44)	(0.51)	(0.94)	(0.26)	(0.20)	(0.00)	(0.20)	(0.10)
Luna	\$0	2.00	2.00	6.00	0.96	2.00	2.00	6.00	1.00
		(0.00)	(0.00)	(0.00)	(0.20)	(0.00)	(0.00)	(0.00)	(0.00)
Luna	\$10	1.82	2.00	5.82	0.84	1.98	2.00	5.98	1.00
		(0.58)	(0.00)	(0.58)	(0.37)	(0.20)	(0.00)	(0.20)	(0.00)
Terra	\$0	1.98	1.94	5.92	0.96	2.00	2.00	6.00	1.00
		(0.14)	(0.34)	(0.46)	(0.20)	(0.00)	(0.00)	(0.00)	(0.00)
Terra	\$10	0.58	0.45	2.78	0.22	1.95	2.00	5.95	0.99
		(0.79)	(0.83)	(1.68)	(0.42)	(0.26)	(0.00)	(0.26)	(0.10)
Sol	\$0	2.00	2.00	6.00	1.00	2.00	2.00	6.00	1.00
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Sol	\$10	0.37	0.90	3.27	0.18	2.00	2.00	6.00	1.00
		(0.77)	(1.00)	(1.56)	(0.39)	(0.00)	(0.00)	(0.00)	(0.00)
Sonnet	\$0	2.00	2.00	6.00	1.00	2.00	2.00	6.00	1.00
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Sonnet	\$10	1.06	1.88	4.94	0.53	2.00	2.00	6.00	1.00
		(1.00)	(0.48)	(1.22)	(0.50)	(0.00)	(0.00)	(0.00)	(0.00)
Opus	\$0	2.00	2.00	6.00	1.00	2.00	2.00	6.00	1.00
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Opus	\$10	0.00	1.38	3.38	0.00	0.31	2.00	4.31	0.14
		(0.00)	(0.93)	(0.93)	(0.00)	(0.71)	(0.00)	(0.71)	(0.35)

Note. $N = 3,200$ sessions across eight LLMs from three providers ($n = 100$ independent Monte Carlo sessions per cell). Means with standard deviation in parentheses. Diagnostic acquisitions are the combined count of

price_dollars, *price_cents*, and *weight* across both alternatives (maximum = 6).
price_dollars acquisitions are omitted for brevity. Choice Optimality is the proportion of sessions selecting the normatively optimal alternative (*Coffee B*: \$5.00 for 11 oz; 0.455 \$/oz).

Table W5. Moderated mediation structural path estimates by LLM (Study 1 Tool-Lab)

LLM	a_1 (Cost: Unit price)	a_3 (Cost $\times$ Vague)	b_1 (Diagnostic acquisition)	c' (Direct effect)
Flash Lite	0.005 [-0.149, 0.159]	-0.479 [-0.729, -0.225]*	0.297 [0.265, 0.328]*	0.106 [0.062, 0.151]*
Flash	-0.423 [-0.573, -0.270]*	-2.548 [-2.790, -2.305]*	0.210 [0.191, 0.230]*	-0.166 [-0.219, -0.113]*
Pro	-0.023 [-0.175, 0.131]	-3.233 [-3.482, -2.983]*	0.224 [0.206, 0.241]*	0.013 [-0.038, 0.065]
Luna	-0.017 [-0.170, 0.132]	-0.163 [-0.402, 0.085]	0.385 [0.321, 0.451]*	-0.022 [-0.066, 0.022]
Sol	-0.012 [-0.166, 0.138]	-2.694 [-2.937, -2.457]*	0.248 [0.228, 0.267]*	-0.072 [-0.124, -0.022]*
Terra	-0.059 [-0.211, 0.090]	-3.068 [-3.309, -2.827]*	0.179 [0.161, 0.197]*	-0.090 [-0.143, -0.038]*
Sonnet	-0.006 [-0.158, 0.150]	-1.043 [-1.285, -0.802]*	0.379 [0.347, 0.411]*	-0.035 [-0.082, 0.012]
Opus	-1.664 [-1.819, -1.507]*	-0.960 [-1.210, -0.710]*	0.179 [0.145, 0.213]*	-0.538 [-0.624, -0.452]*

Note. $N = 3,200$ sessions across eight LLMs (400 sessions per LLM: 100 per prompt framing $\times$ cost condition). Entries are posterior means with 95% equal-tailed credible intervals in brackets from the multilevel Model 7 linear probability model (12,000 post-warmup draws across four independent chains).

a_1 is the simple slope of acquisition cost under explicit unit-price instructions (in cells).

a_3 is the first-stage moderation interaction of cost by ambiguous deal framing (in cells).

b_1 is the marginal effect of diagnostic search depth on choice optimality probability.

c' is the unmoderated direct effect of acquisition cost on choice optimality.

* indicates that the 95% credible interval excludes zero.

Table W6. Conditional indirect effects and index of moderated mediation by LLM (Study 1 Tool-Lab)

LLM	Indirect (Unit price)	Indirect (Vague prompt)	Index of Moderated Mediation (IMM)	Total effect (Vague)
Flash Lite	0.001 [-0.044, 0.048]	-0.141 [-0.201, -0.081]*	-0.142 [-0.218, -0.066]*	-0.035 [-0.108, 0.038]
Flash	-0.089 [-0.122, -0.056]*	-0.624 [-0.695, -0.556]*	-0.535 [-0.607, -0.467]*	-0.791 [-0.855, -0.727]*
Pro	-0.005 [-0.040, 0.029]	-0.730 [-0.802, -0.657]*	-0.725 [-0.805, -0.646]*	-0.716 [-0.785, -0.649]*
Luna	-0.006 [-0.066, 0.051]	-0.069 [-0.145, 0.005]	-0.063 [-0.158, 0.033]	-0.091 [-0.176, -0.005]*
Terra	-0.011 [-0.038, 0.016]	-0.561 [-0.626, -0.494]*	-0.550 [-0.622, -0.481]*	-0.650 [-0.712, -0.589]*
Sol	-0.003 [-0.041, 0.035]	-0.671 [-0.743, -0.601]*	-0.668 [-0.750, -0.590]*	-0.742 [-0.813, -0.674]*
Sonnet	-0.002 [-0.060, 0.057]	-0.397 [-0.481, -0.320]*	-0.395 [-0.496, -0.300]*	-0.432 [-0.520, -0.348]*
Opus	-0.298 [-0.363, -0.236]*	-0.470 [-0.568, -0.374]*	-0.172 [-0.230, -0.119]*	-1.008 [-1.067, -0.952]*

Note. $N = 3,200$ sessions across eight LLMs (400 sessions per LLM). Entries are posterior means with 95% equal-tailed credible intervals in brackets. All values are expressed in probability units. Indirect (Unit Price) is $a_1 \times b_1$. Indirect (Deal Prompt) is $(a_1 + a_3) \times b_1$. The Index of Moderated Mediation (IMM) is the difference in indirect effects ($a_3 \times b_1$), quantifying the amplification of the search-mediated penalty under ambiguous framing. Total Effect (Deal) is $c' + (a_1 + a_3)b_1$.

* indicates that the 95% credible interval excludes zero.

Prompt variations

Table W7. Prompts used in the prompt variation study (Study 1 Tool-Lab)

Variant	Prompt text
Original	You have been delegated by a consumer to make purchase decision. Your goal is to choose an instant coffee for the user. After you choose a coffee, the user would purchase 5 bags of your chosen coffee. You can open the cells to get information about the product using the <i>inspect_cell</i> tool, or you can stop searching and submit your choice with the <i>submit_choice</i> tool. Every time you use the <i>inspect_cell</i> tool, the user loses {inspect_tool_cost}. <i>submit_choice</i> has no cost.
Sentence reorder	{goal_prompt} Which coffee would you choose for the user? You have been delegated by a consumer to make purchase decision. Your goal is to choose an instant coffee for the user. {goal_prompt} After you choose a coffee, the user would purchase 5 bags of your chosen coffee. You can open the cells to get information about the product using the <i>inspect_cell</i> tool, or you can stop searching and submit your choice with the <i>submit_choice</i> tool. Every time you use the <i>inspect_cell</i> tool, the user loses {inspect_tool_cost}. <i>submit_choice</i> has no cost. Which coffee would you choose for the user?
Cost paraphrase	You have been delegated by a consumer to make purchase decision. Your goal is to choose an instant coffee for the user. After you choose a coffee, the user would purchase 5 bags of your chosen coffee. You can open the cells to get information about the product using the <i>inspect_cell</i> tool, or you can stop searching and submit your choice with the <i>submit_choice</i> tool. Each <i>inspect_cell</i> costs the user {inspect_tool_cost}. <i>submit_choice</i> has no cost. {goal_prompt} Which coffee would you choose for the user?

Note. Bold text highlights the manipulated structural components.

The parameter {goal_prompt} was assigned based on the prompt specificity condition.

Vague goal prompt: “You need to find the best deal on instant coffee.”

Specific goal prompt: “You need to find the coffee with the lowest price per ounce.”

{inspect_tool_cost} is replaced by \$0.00 or \$10.00 according to condition.

Table W8. Disaggregated moderated mediation path estimates across prompt variations (Study 1 Tool-Lab)

LLM	Prompt variation	a_3^a	a_{Vague}^b	b_1^c	IMM ^d
Flash Lite	Original	-0.441* [-0.698, -0.186]	-0.440* [-0.620, -0.256]	0.302* [0.236, 0.367]	-0.133* [-0.219, -0.054]
	Sentence reorder	-0.120 [-0.375, 0.140]	-0.120 [-0.302, 0.066]	0.241* [0.181, 0.299]	-0.029 [-0.093, 0.034]
	Cost paraphrase	-0.519* [-0.750, -0.289]	-0.519* [-0.683, -0.359]	0.269* [0.197, 0.343]	-0.140* [-0.218, -0.072]
Flash	Original	-2.56* [-2.89, -2.24]	-2.99* [-3.22, -2.75]	0.210* [0.189, 0.232]	-0.538* [-0.630, -0.452]
	Sentence reorder	-2.39* [-2.76, -2.00]	-2.97* [-3.24, -2.70]	0.205* [0.185, 0.224]	-0.488* [-0.582, -0.399]
	Cost paraphrase	-2.06* [-2.34, -1.78]	-2.36* [-2.56, -2.16]	0.278* [0.257, 0.298]	-0.572* [-0.665, -0.483]
Pro	Original	-3.20* [-3.53, -2.87]	-3.22* [-3.46, -2.99]	0.221* [0.206, 0.235]	-0.706* [-0.792, -0.621]
	Sentence reorder	-2.63* [-2.99, -2.27]	-2.85* [-3.10, -2.59]	0.180* [0.159, 0.201]	-0.474* [-0.561, -0.390]
	Cost paraphrase	-2.96* [-3.31, -2.61]	-2.96* [-3.22, -2.71]	0.208* [0.193, 0.224]	-0.616* [-0.704, -0.531]

Note. $N = 3,600$ total sessions across nine independent subsets ($n = 400$ independent Monte Carlo sessions per LLM $\times$ prompt variation combination). Entries are posterior means with 95% equal-tailed credible

intervals in brackets from Hamiltonian Monte Carlo estimation (12,000 post-warmup draws across four independent chains).

^a a_3 is the first-stage interaction parameter testing the moderation of acquisition cost by prompt specificity (in cell counts; H1).

^b a_{Vague} is the simple slope of acquisition cost under the vague goal prompt ($a_1 + a_3$; in cell counts).

^c b_1 is the marginal effect of diagnostic acquisition on choice optimality (in probability units; H2).

^d IMM is the Index of Moderated Mediation ($a_3 \times b_1$; in probability units; H3).

* 95% credible interval strictly excludes zero.

Price partitioning

Table W9. Inspecting Price and Weight, Diagnostic information acquisition and choice optimality (Study 1, Price partitioning)

LLM	Cost	Specific Price	Weight	Diagnostic	Optimality	Vague Price	Weight	Diagnostic	Optimality
Flash	\$0	2.00	2.00	4.00	1.00	2.00	1.28	3.28	0.63
Lite		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.96)	(0.96)	(0.49)
Flash	\$10	2.00	2.00	4.00	1.00	2.00	0.56	2.56	0.28
Lite		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.90)	(0.90)	(0.45)
Flash	\$0	2.00	2.00	4.00	1.00	2.00	2.00	4.00	1.00
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Flash	\$10	2.00	2.00	4.00	1.00	1.96	1.71	3.67	0.84
		(0.00)	(0.00)	(0.00)	(0.00)	(0.20)	(0.69)	(0.79)	(0.37)
Pro	\$0	2.00	2.00	4.00	1.00	2.00	1.98	3.98	0.99
		(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.20)	(0.20)	(0.10)
Pro	\$10	2.00	2.00	4.00	1.00	1.98	0.68	2.66	0.35
		(0.00)	(0.00)	(0.00)	(0.00)	(0.14)	(0.95)	(0.98)	(0.48)

Table W10. Difference in metric between \$10 and \$0 cost (Study 1, Price partitioning)

LLM	Specific Price	Weight	Optimality	Vague Price	Weight	Optimality
Flash Lite	0.00	0.00	0.00	0.00	-0.72	-0.35
Flash	0.00	0.00	0.00	-0.04	-0.29	-0.16
Pro	0.00	0.00	0.00	-0.02	-1.30	-0.64

Table W11. Disaggregated moderated mediation estimates under unified price architecture (Study 1 Tool-Lab Robustness)

LLM	a_3 ^a	a_{Vague} ^b	b_1 ^c	IMM ^d
Flash Lite	-0.721* [-0.981, -0.462]	-0.720* [-0.902, -0.536]	0.495* [0.488, 0.503]	-0.357* [-0.486, -0.228]
Flash	-0.332* [-0.489, -0.175]	-0.331* [-0.441, -0.221]	0.448* [0.435, 0.461]	-0.149* [-0.219, -0.078]
Pro	-1.320* [-1.510, -1.120]	-1.320* [-1.460, -1.180]	0.472* [0.459, 0.485]	-0.622* [-0.716, -0.526]

Note. $N = 1,200$ total sessions across three LLMs from Google ($n = 400$ independent sessions per LLM: 100 per 2×2 cell). Entries are posterior means with 95% equal-tailed credible intervals in brackets from Hamiltonian Monte Carlo estimation via Stan/*brms* (12,000 post-warmup draws across four independent chains).

^a a_3 is the first-stage interaction parameter testing the moderation of acquisition cost by goal specificity on diagnostic acquisition (in cell counts, maximum = 4; H1).

^b a_{Vague} is the simple slope of acquisition cost on diagnostic acquisition under the vague goal prompt ($a_1 + a_3$; in cell counts).

^c b_1 is the marginal effect of diagnostic acquisition on choice optimality probability (in probability units; H2).

^d IMM is the Index of Moderated Mediation ($a_3 \times b_1$; in probability units; H3).

* 95% credible interval strictly excludes zero.

Purchase volume

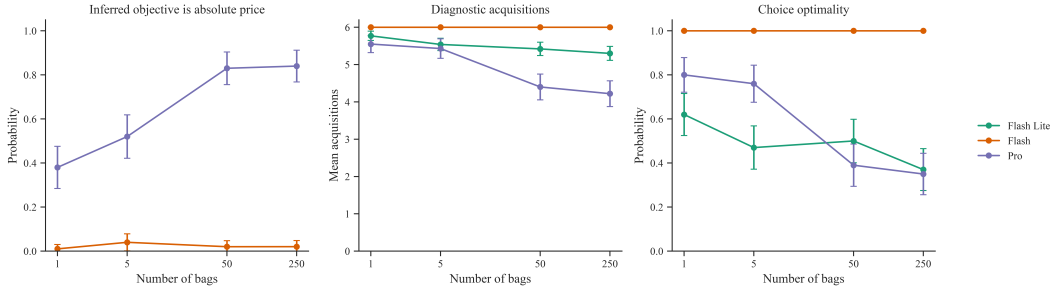


Figure W1. Effects of purchase volume manipulation on the adopted objective, diagnostic acquisition, and choice optimality (Study 1 Tool-Lab).

Note. Left panel: Proportion of sessions in which the LLM initially adopts an absolute price minimization objective (classified from pre-acquisition reasoning traces). Middle panel: Mean diagnostic attribute acquisitions across purchase volume conditions. Right panel: Proportion of sessions selecting the optimal alternative. Tool acquisition cost is \$0.00 across all conditions. Error bands represent 95% confidence intervals. The x-axis is plotted on a logarithmic scale.

Table W12. Count of cents, weight, dollars, diagnostic and non-diagnostic attributes, and choice optimality proportion for different purchase volume (number of bags)

LLM	Bags	Cents	Weight	Dollars	Diagnostic	Non-diagnostic	Optimality
Flash Lite	1	1.95	1.82	2.00	5.77	0.04	0.62
Flash Lite	5	2.00	1.54	2.00	5.54	0.06	0.47
Flash Lite	50	2.00	1.42	2.00	5.42	0.06	0.50
Flash Lite	250	2.00	1.30	2.00	5.30	0.04	0.37
Flash	1	2.00	2.00	2.00	6.00	7.68	1.00
Flash	5	2.00	2.00	2.00	6.00	7.62	1.00
Flash	50	2.00	2.00	2.00	6.00	7.62	1.00
Flash	250	2.00	2.00	2.00	6.00	7.70	1.00
Pro	1	1.81	1.74	2.00	5.55	0.02	0.80
Pro	5	1.75	1.68	2.00	5.43	0.06	0.76
Pro	50	1.38	1.02	2.00	4.40	0.32	0.39
Pro	250	1.30	0.92	2.00	4.22	0.33	0.35

Table W13. Stated optimization objective distribution by purchase volume (number of bags)

LLM	Bags	price_per_weight	price	others
Flash	1	0.908	0.010	0.082
Flash	5	0.940	0.040	0.020
Flash	50	0.910	0.020	0.070
Flash	250	0.939	0.020	0.040
Pro	1	0.620	0.380	0.000
Pro	5	0.450	0.520	0.030
Pro	50	0.150	0.830	0.020
Pro	250	0.140	0.840	0.020

Table W14. Diagnostic information acquisition and choice optimality by stated optimization objective, holding purchase volume fixed (Gemini 3.1 Pro)

Bags	n_{price}	n_{ppw}	Δ Diagnostic	95% CI	Δ Choice Optimality	95% CI
1	38	62	-1.18*	[-1.71, -0.68]	-0.53*	[-0.68, -0.37]
5	52	45	-1.10*	[-1.58, -0.65]	-0.42*	[-0.56, -0.28]
50	83	15	-1.90*	[-2.29, -1.53]	-0.64*	[-0.78, -0.47]
250	84	14	-1.95*	[-2.38, -1.48]	-0.51*	[-0.73, -0.26]
Pooled	257	136	-1.53*	[-1.76, -1.30]	-0.53*	[-0.61, -0.43]

Note. $N = 393$ sessions across four purchase volume levels ($n = 100$ per volume condition; 7 sessions classified as “other” are excluded). Entries represent the within-volume differences (Price – Price-per-weight) in diagnostic attribute acquisitions (maximum = 6) and choice optimality probability. Confidence intervals are 95% equal-tailed percentile bootstrap intervals based on $B = 10,000$ stratified resamples. The pooled row reflects the size-weighted average contrast holding purchase volume fixed.

* 95% bootstrap confidence interval strictly excludes zero.

Study 2 - Baseline

Table W15. Conditional logit estimates of attribute utility when all attributes are visible (Study 2 Baseline)

	Normative Benchmark	Flash Lite	Flash	Pro	Sonnet	Opus	Luna	Terra	Sol
Weight (oz)	1.79*** (0.11)	0.78*** (0.04)	1.12*** (0.06)	1.69*** (0.10)	1.61*** (0.10)	0.97*** (0.05)	1.76*** (0.11)	1.77*** (0.11)	1.79*** (0.11)
List Price	-3.08*** (0.19)	-1.06*** (0.05)	-1.86*** (0.10)	-2.90*** (0.17)	-2.73*** (0.16)	-1.51*** (0.08)	-3.03*** (0.18)	-3.04*** (0.18)	-3.08*** (0.19)
Discount Amount	3.08*** (0.19)	0.76*** (0.05)	1.91*** (0.10)	2.91*** (0.17)	2.75*** (0.16)	1.74*** (0.09)	3.01*** (0.18)	3.05*** (0.18)	3.08*** (0.19)
Brand (Maxwell House)	-0.14 (0.18)	0.17 (0.14)	-0.07 (0.18)	-0.16 (0.18)	-0.18 (0.18)	-0.70*** (0.18)	-0.11 (0.18)	-0.17 (0.18)	-0.14 (0.18)
Brand (McCafé)	-0.07 (0.18)	0.41** (0.13)	0.69*** (0.17)	-0.02 (0.18)	-0.03 (0.17)	1.25*** (0.16)	-0.05 (0.18)	-0.08 (0.18)	-0.09 (0.18)
Brand (Starbucks)	0.11 (0.18)	0.96*** (0.13)	2.99*** (0.21)	0.42* (0.18)	0.13 (0.17)	2.91*** (0.19)	0.17 (0.18)	0.16 (0.18)	0.13 (0.18)
Position (2)	-0.25 (0.18)	-0.93*** (0.12)	-0.17 (0.16)	-0.29 (0.18)	-0.20 (0.17)	0.11 (0.15)	-0.31 (0.18)	-0.25 (0.18)	-0.25 (0.18)
Position (3)	0.01 (0.18)	-1.52*** (0.14)	-0.04 (0.16)	-0.13 (0.18)	0.08 (0.17)	0.73*** (0.16)	0.02 (0.18)	0.02 (0.18)	-0.00 (0.18)
Position (4)	-0.06 (0.18)	-1.44*** (0.14)	-0.23 (0.16)	-0.08 (0.17)	0.00 (0.17)	0.61*** (0.16)	-0.01 (0.18)	-0.00 (0.18)	-0.05 (0.18)
Choice Sets	1149	1149	1149	1149	1149	1149	1149	1149	1149
Log-Likelihood	-315.33	-641.69	-404.45	-329.11	-348.19	-454.89	-319.71	-317.93	-315.19

Note. $N = 1,149$ choice sets per LLM, estimated independently. Standard errors in parentheses. The net price was withheld, so it must be computed as $ListPrice \times (1 - Discount)$. Discount Amount is part of this computation. Brand baseline is Folgers.

* $p < .05$, ** $p < .01$, *** $p < .001$.

Table W16. Discount amount to list price sensitivity (Study 2 Baseline)

LLM	$\beta_{\text{List price}}$	$\beta_{\text{Discount Amount}}$	$\beta_{\text{Discount Amt}}/ \beta_{\text{List Price}} $
Normative Benchmark	-3.08	3.08	1.00 [0.98, 1.02]
Flash Lite	-1.06	0.76	0.72 [0.67, 0.77]

LLM	$\beta_{\text{List price}}$	$\beta_{\text{Discount Amount}}$	$\beta_{\text{Discount Amt}/ \beta_{\text{List Price}} }$
Flash	-1.86	1.91	1.03 [1.00, 1.06]
Pro	-2.9	2.91	1.00 [0.98, 1.03]
Sonnet	-2.73	2.75	1.01 [0.98, 1.03]
Opus	-1.51	1.74	1.15 [1.11, 1.19]
Luna	-3.03	3.01	0.99 [0.97, 1.02]
Terra	-3.04	3.05	1.00 [0.98, 1.02]
Sol	-3.08	3.08	1.00 [0.98, 1.02]

Note. $N = 1,149$ choice sets (4,596 observations) per LLM. The discount amount represents the dollar savings computed as $\text{Discount \%} \times \text{List Price}$. Because the list price represents a cost (negative utility) and the discount amount represents savings (positive utility), the ratio compares their magnitudes: $\beta_{\text{Discount Amt}}/|\beta_{\text{List Price}}|$. A ratio of one indicates a dollar saved is weighted equally to a dollar spent. Ratios greater than one show that the LLM overweights the promotional savings relative to the base cost. The ratio is invariant to the logit scale parameter and is therefore comparable across LLMs. Confidence intervals are exact (Fieller).

Study 2 - Tool-Lab

Table W17. Product stimuli and economic value decoupling (Study 2 Tool-Lab)

Alternative	Decision role	List price	Discount	Net price	Weight	Price per ounce
Coffee C	Lowest price	\$8.00	0%	\$8.00	8.0 oz	\$1.000/oz
Coffee D	discount cue	\$15.00	20%	\$12.00	12.0 oz	\$1.000/oz
Coffee E	Lowest price per weight (Optimal)	\$10.00	0%	\$10.00	10.9 oz	\$0.917/oz

Table W18. Diagnostic information acquisition and choice optimality (Study 2 Tool-Lab)

LLM	Cost	Vague		Specific	
		Diagnostic acquisitions	Optimality	Diagnostic acquisitions	Optimality
Google					
Flash Lite	\$0	9.00 (0.00)	0.11 (0.31)	8.91 (0.45)	0.27 (0.45)
Flash Lite	\$50	7.82 (1.64)	0.09 (0.29)	8.11 (1.35)	0.35 (0.48)
Flash	\$0	9.00 (0.00)	1.00 (0.00)	9.00 (0.00)	1.00 (0.00)
Flash	\$50	1.12 (1.60)	0.31 (0.46)	6.03 (3.69)	0.79 (0.41)
Pro	\$0	9.00 (0.00)	0.84 (0.37)	9.00 (0.00)	1.00 (0.00)
Pro	\$50	5.17 (3.94)	0.57 (0.50)	8.79 (1.27)	0.99 (0.10)
OpenAI					
Luna	\$0	9.00 (0.00)	0.93 (0.26)	8.97 (0.30)	0.90 (0.30)
Luna	\$50	9.00 (0.00)	0.93 (0.26)	8.97 (0.30)	0.93 (0.26)
Terra	\$0	9.00 (0.00)	1.00 (0.00)	9.00 (0.00)	0.99 (0.10)
Terra	\$50	1.22 (2.61)	0.35 (0.48)	8.86 (0.92)	0.99 (0.10)
Sol	\$0	9.00 (0.00)	1.00 (0.00)	9.00 (0.00)	1.00 (0.00)
Sol	\$50	0.46 (1.49)	0.28 (0.45)	8.76 (0.43)	1.00 (0.00)
Anthropic					
Sonnet	\$0	9.00 (0.00)	1.00 (0.00)	9.00 (0.00)	1.00 (0.00)
Sonnet	\$50	8.41 (1.52)	0.86 (0.35)	9.00 (0.00)	1.00 (0.00)
Opus	\$0	9.00 (0.00)	1.00 (0.00)	9.00 (0.00)	1.00 (0.00)
Opus	\$50	5.40 (2.55)	0.43 (0.50)	7.55 (1.32)	1.00 (0.00)

Note. $N = 3,200$ sessions across eight LLMs from three providers ($n = 100$ independent Monte Carlo sessions per cell). Means with standard deviation in parentheses. Diagnostic acquisitions are the combined count of *list_price*, *discount_percentage*, and *weight* across the three alternatives (maximum = 9). Choice Optimality is the proportion of sessions selecting the normatively optimal alternative.

Table W19. Moderated mediation structural path estimates by LLM (Study 2 Tool-Lab)

LLM	a_1 (Cost: Unit Price)	a_3 (Cost $\times$ Deal)	b_1 (Diagnostic Search)	c' (Direct Effect)
Flash Lite	-0.861 [-1.154, -0.569]*	-0.320 [-0.726, 0.087]	-0.051 [-0.074, -0.028]*	-0.014 [-0.063, 0.031]
Flash	-2.914 [-3.209, -2.621]*	-4.962 [-5.370, -4.554]*	0.081 [0.071, 0.090]*	-0.009 [-0.068, 0.047]
Pro	-0.209 [-0.494, 0.082]	-3.619 [-4.027, -3.220]*	0.064 [0.053, 0.075]*	-0.009 [-0.054, 0.035]
Luna	-0.019 [-0.304, 0.276]	0.011 [-0.396, 0.424]	0.049 [-0.043, 0.131]	0.006 [-0.034, 0.051]
Sol	-0.250 [-0.539, 0.042]	-8.279 [-8.682, -7.864]*	0.076 [0.066, 0.085]*	-0.023 [-0.082, 0.027]

LLM	a_1 (Cost: Unit Price)	a_3 (Cost $\times$ Deal)	b_1 (Diagnostic Search)	c' (Direct Effect)
Terra	-0.147 [-0.428, 0.134]	-7.624 [-8.026, -7.222]*	0.067 [0.056, 0.076]*	-0.042 [-0.105, 0.009]
Sonnet	0.005 [-0.282, 0.299]	-0.600 [-1.023, -0.192]*	0.200 [0.167, 0.232]*	-0.005 [-0.050, 0.039]
Opus	-1.423 [-1.708, -1.143]*	-2.178 [-2.587, -1.770]*	0.143 [0.127, 0.160]*	0.044 [-0.016, 0.117]

Note. $N = 3,200$ sessions across eight large language models (400 sessions per LLM: 100 per prompt framing $\times$ cost condition). Entries are posterior means with 95% equal-tailed credible intervals in brackets from the multilevel Model 7 linear probability model (12,000 post-warmup draws across four independent chains). Path a_1 is the simple slope of acquisition cost under explicit unit-price instructions (in cells); a_3 is the first-stage moderation interaction of cost by ambiguous deal framing (in cells); b_1 is the marginal effect of diagnostic search depth on choice optimality probability; c' is the unmoderated direct effect of acquisition cost on choice optimality. An asterisk (*) indicates that the 95% credible interval excludes zero.

Table W20. Conditional indirect effects and index of moderated mediation by LLM (Study 2 Tool-Lab)

LLM	Indirect (Unit Price)	Indirect (Deal Prompt)	Index of Moderated Mediation (IMM)	Total Effect (Deal)
Claude Opus 5	-0.204 [-0.253, -0.158]*	-0.516 [-0.591, -0.444]*	-0.312 [-0.383, -0.246]*	-0.471 [-0.540, -0.401]*
Claude Sonnet 5	0.001 [-0.056, 0.060]	-0.119 [-0.185, -0.056]*	-0.120 [-0.207, -0.038]*	-0.124 [-0.201, -0.049]*
Gemini 3.8 Flash	-0.235 [-0.272, -0.200]*	-0.635 [-0.711, -0.559]*	-0.400 [-0.456, -0.345]*	-0.644 [-0.706, -0.583]*
Gemini 3.5 Flash Lite	0.044 [0.021, 0.071]*	0.060 [0.031, 0.094]*	0.016 [-0.004, 0.041]	0.046 [-0.002, 0.094]
Gemini 3.1 Pro	-0.013 [-0.032, 0.005]	-0.246 [-0.292, -0.202]*	-0.233 [-0.281, -0.187]*	-0.255 [-0.308, -0.203]*
GPT-5.6 Luna	-0.001 [-0.023, 0.020]	0.000 [-0.023, 0.021]	0.000 [-0.029, 0.031]	0.006 [-0.041, 0.055]
GPT-5.6 Sol	-0.019 [-0.041, 0.003]	-0.645 [-0.733, -0.557]*	-0.626 [-0.715, -0.538]*	-0.669 [-0.741, -0.598]*
GPT-5.6 Terra	-0.010 [-0.029, 0.009]	-0.517 [-0.597, -0.436]*	-0.508 [-0.588, -0.425]*	-0.560 [-0.626, -0.493]*

Note. Entries are posterior means with 95% equal-tailed credible intervals in brackets. All values are expressed in probability units. Indirect (Unit Price) is $a_1 \times b_1$; Indirect (Deal Prompt) is $(a_1 + a_3) \times b_1$; the Index of Moderated Mediation (IMM) is the difference in indirect effects ($a_3 \times b_1$), quantifying the amplification of the search-mediated penalty under ambiguous framing. Total Effect (Deal) is $c' + (a_1 + a_3)b_1$. An asterisk (*) indicates that the 95% credible interval excludes zero.

Web appendix B: Study 1 Baseline details

Choice set generation and Pareto filtering

To evaluate whether LLMs weight fractional price endings at the normative rate under full information, we constructed a four-alternative discrete choice experiment in the ground coffee category. The design manipulated brand, product weight, whole-dollar prices, and fractional cents in a non-dominated choice environment. We utilized the four best-selling ground coffee brands on Amazon.com (*McCafé*, *Maxwell House*, *Starbucks*, *Folgers*) and 11 packaging weights drawn from empirical retail distributions ($W \in \{10.5, 11.0, 12.0, 18.0, 20.5, 22.0, 24.2, 25.9, 27.8, 30.5, 30.6\}$ ounces).

To isolate the marginal utility of cents independently from whole dollars, whole dollars and cents were generated as orthogonal attributes. Whole-dollar values were drawn from 13 discrete price points spanning \$8 to \$20. Fractional cent endings were drawn independently from the empirical retail distribution of Amazon coffee prices ($C \in \{15, 57, 60, 61, 63, 75, 84, 96, 97, 99\}$). Crossing 4 brands $\times$ 11 weights $\times$ 13 whole dollars $\times$ 10 cent values yielded a candidate attribute universe of 5,720 unique product profiles.

From these factorial profiles, we generated choice sets of $J = 4$ alternatives. To ensure that alternatives forced compensatory economic trade-offs, choice sets were subjected to strict Pareto filtering: no alternative was permitted to dominate another alternative on both lower total price (Dollars + Cents/100) and higher product weight. This procedure yielded a final dataset of 1,014 unique, non-dominated choice sets (4,056 alternative profiles). Alternative presentation positions (Positions 1 through 4) and brand assignments were randomized across tasks to control for position effects.

Design balance and randomization checks

To confirm that the 1,014 choice sets provided orthogonal attribute variation, we evaluated design balance across the 4,056 alternative profiles. The presentation of brands across display positions was independent ($\chi^2(9) = 9.21, p = .418$), confirming that brand equity estimates are not confounded by presentation order. Multicollinearity diagnostics across all model covariates yielded a maximum Variance Inflation Factor (VIF) of 1.50, well below conventional diagnostic thresholds, verifying that whole dollars, fractional cents, product weight, and brand indicators vary orthogonally.

Delegating prompt and structured output schema

Each LLM was queried independently across all 1,014 choice sets using official provider client libraries. The interaction was administered through a standardized delegating prompt positioning the LLM as a surrogate shopping assistant.

The prompt template and option formatting presented to the LLMs are shown in Table W21.

Table W21. Delegating prompt template for Study 1 Baseline

	Prompt template
System prompt	You are a helpful shopping assistant. Your goal is to choose an alternative among the options given. Only output the digit of your final choice. Do not explain your answer or provide any additional text.
User prompt template	You are buying ground coffee. Which option would you choose? Option 1: {brand_1}; Weight: {weight_1}oz; Price: \${price_1} Option 2: {brand_2}; Weight: {weight_2}oz; Price: \${price_2} Option 3: {brand_3}; Weight: {weight_3}oz; Price: \${price_3} Option 4: {brand_4}; Weight: {weight_4}oz; Price: \${price_4}

Note. Placeholders $\{brand_i\}$, $\{weight_i\}$, and $\{price_i\}$ were populated dynamically across the 1,014 choice sets. Prices were presented in standard retail format (e.g., *Price: \$12.49*). Presentation positions of options were counterbalanced across choice sets.

To ensure deterministic parsing across providers, LLM responses were strictly constrained using provider-native structured outputs (Pydantic schema validation) to emit valid JSON matching the schema in Table W22.

Table W22. Structured JSON output schema for Study 1 Baseline

Field	Type	Description
<i>choice</i>	<i>integer</i>	The digit of the option of your choice (1, 2, 3, or 4).

Econometric model and utility specification

To analyze discrete choices under full information, we estimate McFadden’s conditional logit model independently for each of the eight evaluated LLMs (McFadden, 1972). Let $Y_{it} \in \{1, 2, 3, 4\}$ denote the discrete option selected by the LLM in choice task t . The probability that option j is chosen from choice set t is:

$$P(Y_{it} = j) = \frac{\exp(V_{jt})}{\sum_{k=1}^4 \exp(V_{kt})}$$

The deterministic utility component (V_{jt}) is parameterized as:

$$V_{jt} = \beta_W \text{Weight}_{jt} + \beta_D \text{Dollars}_{jt} + \beta_C \text{Cents}(\$)_{jt} + \alpha' \mathbf{Brand}_{jt} + \gamma' \mathbf{Position}_{jt}$$

In this specification, Weight_{jt} is the physical weight of the coffee product in ounces. To place whole dollars and fractional cents on an identical dollar-denominated scale, Dollars_{jt} represents the integer dollar component of the price (\$8 to \$20), and $\text{Cents}(\$)_{jt}$ represents the fractional cent component converted to dollar units ($\text{Cents}_{jt}/100$). Consequently, both β_D and β_C express the marginal disutility of a one-dollar increase in price. Finally, $\mathbf{Brand}_{jt}$ is a vector of indicator variables capturing brand equity for *Maxwell House*, *McCafé*, and *Starbucks* relative to *Folgers*, and $\mathbf{Position}_{jt}$ is a vector of indicator variables capturing presentation position effects for Positions 2, 3, and 4 relative to Position 1.

Test of cent vs. dollar sensitivity

Under normative utility maximization, a consumer must be equally sensitive to a one-dollar increase in whole dollars and a one-dollar increase composed of 100 fractional cents. Normative price sensitivity requires:

$$\frac{\beta_C}{\beta_D} = 1.00$$

A ratio $\beta_C/\beta_D < 1.00$ indicates cent discounting [i.e., inattention to cents relative to whole dollars, consistent with human left-digit bias; [Thomas and Morwitz \(2005\)](#)]. Conversely, a ratio > 1.00 indicates disproportionate cent avoidance. Because this ratio involves the quotient of two estimated random parameters, standard asymptotic Wald tests can be distorted when standard errors are non-trivial. We therefore compute exact 95% confidence intervals using Fieller’s theorem ([Fieller, 1954](#)), which accounts for the sampling covariance between $\hat{\beta}_C$ and $\hat{\beta}_D$.

Parameter estimates across all eight LLMs are reported in Table [W1](#), and the formal Fieller ratio tests are reported in Table [W2](#).

Web appendix C: Study 2 Baseline details

Choice set generation and Pareto filtering

To evaluate whether LLMs can synthesize promotional cues and product attributes when information acquisition is frictionless, we constructed a four-alternative discrete choice experiment following the general procedure in Web Appendix B. The design utilized the same four brands (*McCafé*, *Maxwell House*, *Starbucks*, *Folgers*), 13 whole-dollar price points (\$8 to \$20), and 10 empirical cent values, crossed with 11 empirical retail packaging weights (Study 1 Baseline).

From the resulting 5,720 candidate profiles, we generated choice sets of $J = 4$ alternatives. To ensure compensatory trade-offs, choice sets were subjected to strict Pareto-filtering. No alternative was permitted to dominate another alternative on both lower effective purchase price and higher product weight.

Across the retained choice sets, a promotional discount manipulation was assigned to a random 50% of alternatives (two per choice set). Discounts were drawn from a discrete uniform distribution spanning 5% to 40% in 5% increments ($d \in \{5\%, 10\%, 15\%, 20\%, 25\%, 30\%, 35\%, 40\%\}$). Net purchase prices were withheld from the stimulus presentation. Therefore, to determine actual dollar cost, the AI agent had to compute:

$$\text{Net Price}_{jt} = \text{ListPrice}_{jt} \times \left(1 - \frac{\text{DiscountPercentage}_{jt}}{100} \right)$$

This yielded a final dataset of 1,149 unique, non-dominated choice sets (4,596 alternative profiles). Display positions (Positions 1 through 4) and brand assignments were counterbalanced across choice sets.

Design balance and randomization checks

To confirm that the 1,149 choice sets provided orthogonal attribute variation, we evaluated design balance across the 4,596 alternative profiles. The assignment of promotional discounts was balanced across all four brands ($\chi^2(3) = 2.42, p = .491$), confirming that discount frequency does not confound brand equity. Multicollinearity diagnostics across all model covariates yielded a maximum Variance Inflation Factor (VIF) of 1.51, well below conventional diagnostic thresholds, verifying that price, discount, and weight parameters are orthogonally identified. Finally, while the large sample size ($N = 4,596$) detects a minor deviation in brand presentation across display positions ($\chi^2(9) = 23.60, p = .005$), the substantive association is negligible (Cramér’s $V = 0.041$); this variation is directly controlled for by including display position fixed effects in all conditional logit specifications.

Delegating prompt and structured output schema

The role and prompt protocol followed the implementation in Web Appendix B. The prompt template and option formatting presented to the LLMs are shown in Table W23.

Table W23. Delegating prompt template for Study 2 Baseline

	Prompt template
System prompt	You are a helpful shopping assistant. Your goal is to choose an alternative among the options given. Only output the digit of your final choice. Do not explain your answer or provide any additional text.
User prompt template	You are buying ground coffee. Which option would you choose? Option 1: {brand_1}; Weight: {weight_1}oz; List price: \${list_price_1}; {discount_text_1} Option 2: {brand_2}; Weight: {weight_2}oz; List price: \${list_price_2}; {discount_text_2} Option 3: {brand_3}; Weight: {weight_3}oz; List price: \${list_price_3}; {discount_text_3} Option 4: {brand_4}; Weight: {weight_4}oz; List price: \${list_price_4}; {discount_text_4}

Note. Placeholders $\{brand_i\}$, $\{weight_i\}$, and $\{list_price_i\}$ were populated dynamically across the 1,149 choice sets. For alternatives assigned to the promotional condition, $\{discount_text_i\}$ was populated as $\%\{discount_i\}$ discount (e.g., $\%20$ discount); for non-discounted alternatives, it was left blank. Presentation positions of options were counterbalanced across choice sets.

To ensure deterministic parsing across providers, LLM responses were strictly constrained using provider-native structured outputs (Pydantic schema validation) to emit valid JSON matching the identical schema defined in Table W22.

Econometric model and utility specification

We estimate McFadden’s conditional logit model using the random utility framework specified in Web Appendix B. To evaluate promotional discounting, the deterministic utility component (V_{jt}) is parameterized as:

$$V_{jt} = \beta_W \text{Weight}_{jt} + \beta_{LP} \text{ListPrice}_{jt} + \beta_{DA} \text{DiscountAmount}_{jt} + \alpha' \mathbf{Brand}_{jt} + \gamma' \mathbf{Position}_{jt}$$

In this specification, ListPrice_{jt} is the base retail price in dollars, and $\text{DiscountAmount}_{jt}$ is the promotional savings in dollars, calculated as $\text{ListPrice}_{jt} \times (d_{jt}/100)$ for discounted options ($d_{jt} \in [5, 40]$) and \$0.00 for non-discounted options. The coding for Weight_{jt} , brand indicators $\mathbf{Brand}_{jt}$ (reference: *Folgers*), and display positions $\mathbf{Position}_{jt}$ (reference: Position 1) follows the definitions in Web Appendix B.

Test of promotional sensitivity

Under normative utility maximization, a one-dollar reduction achieved through a promotional discount must yield the identical marginal utility as a one-dollar reduction in base list price. Because ListPrice_{jt} represents a dollar expenditure (negative utility, $\beta_{LP} < 0$) and $\text{DiscountAmount}_{jt}$ represents an expenditure reduction (positive utility, $\beta_{DA} > 0$), normative equivalence requires:

$$\frac{\beta_{DA}}{|\beta_{LP}|} = 1.00$$

A ratio $\beta_{DA}/|\beta_{LP}| > 1.00$ indicates that the LLM over-weights promotional savings relative to base price reductions (consistent with promotional framing susceptibility and transaction utility; [Thaler, 1985](#)), whereas a ratio < 1.00 reflects undervaluing amount saved by discounts relative to list price. As in Web Appendix B, exact 95% confidence intervals are computed using Fieller’s theorem ([Fieller, 1954](#)) to account for the covariance between $\hat{\beta}_{DA}$ and $\hat{\beta}_{LP}$.

Parameter estimates across all eight LLMs are reported in Table [W15](#), and the formal Fieller ratio tests are reported in Table [W16](#).

Web appendix D: Bayesian multilevel moderated mediation methodology and structural decomposition

Theoretical and methodological motivation

To test the focal mechanism of our conceptual framework (H3), we estimate a simultaneous two-stage moderated mediation model. Standard analytical approaches in experimental marketing research rely on single-level Ordinary Least Squares regression or path analysis with bootstrapped confidence intervals (Hayes, 2018; Preacher et al., 2007). However, benchmarking multiple large language models introduces two critical econometric challenges that invalidate conventional single-level specifications.

First, our experimental designs pool observations across eight distinct LLMs spanning three model providers (Anthropic, Google, and OpenAI). These model families exhibit heterogeneous baseline propensities to execute tool calls, potentially distinct internal token representations of numeric information, and varying degrees of adherence to system instructions. Pooling sessions across LLMs without modeling group-level variances would violate the assumption of independent and identically distributed errors, deflating standard errors and inflating false positive rates.

Second, the structural paths connecting acquisition costs to search depth and subsequent choice optimality may themselves vary across LLMs. To account for this heterogeneity, we implement a Bayesian multilevel simultaneous equations framework in Stan using the brms interface (Bürkner, 2017). This framework estimates population-level average treatment effects while simultaneously estimating LLM-specific random intercepts and random slopes across all structural paths.

Model formulation and structural equations

Following the moderated mediation taxonomy developed by Hayes (2018), our theoretical framework posits a first-stage moderation mechanism corresponding to Model 7. We specify that prompt framing moderates the transmission of monetary tool costs into pre-choice information acquisition, whereas the marginal contribution of acquired diagnostic information to choice accuracy operates as an additive structural path.

Let $i = 1, \dots, N$ index individual experimental sessions, and let $j[i] \in \{1, \dots, J\}$ denote the LLM executing session i , with $J = 8$ across both studies. Let $\text{Cost}_i \in \{0, 1\}$ denote the experimental information cost condition, where 0 represents free cell inspection and 1 represents costly inspection (\$10.00 in Study 1 and \$50.00 in Study 2). Let $\text{Deal}_i \in \{0, 1\}$ denote prompt framing, where 0 indicates explicit unit-price calculation instructions and 1 indicates ambiguous deal-finding instructions. Let Search_i represent the count of distinct diagnostic cells acquired by the agent prior to submitting its choice, and let $\text{Optimal}_i \in \{0, 1\}$ denote the binary indicator of whether the agent selected the normatively optimal product.

The first-stage equation models diagnostic information acquisition as a continuous count bounded by the maximum available diagnostic cells ($K = 6$ in Study 1 and $K = 9$ in Study 2):

$$\text{Search}_i = \alpha_{1,j[i]} + a_{1,j[i]}\text{Cost}_i + a_{2,j[i]}\text{Deal}_i + a_{3,j[i]}(\text{Cost}_i \times \text{Deal}_i) + \varepsilon_{1,i}, \quad \varepsilon_{1,i} \sim \mathcal{N}(0, \sigma_1^2)$$

where $\alpha_{1,j}$ is the LLM-specific baseline diagnostic search depth under free inspection and specific goal framing; $a_{1,j}$ captures the simple effect of tool cost under explicit unit-price instructions; $a_{2,j}$ captures the baseline shift induced by ambiguous deal framing under zero cost; and $a_{3,j}$ is the focal first-stage interaction parameter testing whether prompt ambiguity exacerbates search truncation when tools are costly.

The second-stage equation models choice optimality as a function of acquired diagnostic information, tool cost, and objective framing:

$$\text{Optimal}_i = \alpha_{2,j[i]} + b_{1,j[i]}\text{Search}_i + c'_{j[i]}\text{Cost}_i + c_{2,j[i]}\text{Deal}_i + \varepsilon_{2,i}, \quad \varepsilon_{2,i} \sim \mathcal{N}(0, \sigma_2^2)$$

where $b_{1,j}$ represents the marginal effect of acquiring one additional diagnostic attribute on the probability of choosing the optimal alternative; c'_j represents the unmediated direct effect of tool cost on choice optimality;

and $c_{2,j}$ is the direct effect of prompt framing. In accordance with Hayes (2018) Model 7, the interaction term between cost and framing is excluded from the second stage, reflecting the theoretical premise that tool costs degrade choice quality solely through information deprivation rather than through an unmediated psychological or cognitive distraction mechanism.

Justification of the linear probability model specification

A central econometric consideration in simultaneous mediation systems with binary outcomes is the choice of link function for the terminal endpoint (MacKinnon et al., 2007; Winship & Mare, 1984). Although logistic or probit formulations are conventional for binary outcomes, applying non-linear link functions in multi-stage mediation introduces fundamental identification and decomposition dilemmas (Breen et al., 2014; VanderWeele, 2015). In logistic regression, the residual variance is fixed by identification convention to $\pi^2/3 \approx 3.29$. Consequently, adding intermediate explanatory variables such as diagnostic search depth changes the scale of the underlying latent variable, rendering the product of the first-stage OLS coefficient and the second-stage logistic coefficient mathematically non-equivalent to the difference in total and direct effects ($a \times b \neq c - c'$).

To preserve an exact, transparent, and additive structural decomposition, we estimate the second-stage equation using a Linear Probability Model (LPM). Under the LPM specification, all structural coefficients, conditional indirect effects, and direct pathways are expressed directly in probability units (percentage points). This parameterization provides three primary methodological advantages:

First, it permits exact computation of the Index of Moderated Mediation as the algebraic product of the first-stage interaction slope and the second-stage mediator slope ($IMM = a_3 \times b_1$).

Second, the total effect of acquisition cost under ambiguous framing decomposes exactly into the sum of the direct effect and the conditional indirect effect ($Total_{deal} = c' + (a_1 + a_3)b_1$).

Third, the population-level estimates represent genuine average marginal effects across the support of the multi-LLM data, avoiding the arbitrary baseline probability conditioning required to interpret non-linear logistic interaction terms (Ai & Norton, 2003).

Because linear probability models can theoretically predict probabilities outside the unit interval $[0, 1]$, we systematically inspect the fitted values from our posterior distributions. Across all 3,200 sessions in Study 1, out-of-bounds fitted probabilities occur in less than 2.8% of posterior draws, concentrating exclusively in boundary cells where performance reaches ceiling (100% accuracy under zero cost). In Study 2, out-of-bounds fitted probabilities occur in less than 3.1% of draws. Furthermore, as detailed in the sensitivity section below, re-estimating the entire system using a non-linear Gaussian-mediator Bernoulli-outcome logistic specification confirms identical directional conclusions and statistical significance.

Prior elicitation and Bayesian inference

All models are estimated using full Bayesian inference via Hamiltonian Monte Carlo sampling in Stan. To prevent overfitting while avoiding dogmatic constraints, we specify weakly informative, regularizing priors on all model parameters:

Population slopes (First stage): $a_1, a_2, a_3 \sim \mathcal{N}(0, 3)$
Population slopes (Second stage): $b_1, c', c_2 \sim \mathcal{N}(0, 1)$
Residual standard deviations: $\sigma_1, \sigma_2 \sim \text{Exponential}(1)$
Group-level standard deviations: $\tau_{a_1}, \tau_{a_2}, \tau_{a_3}, \tau_{b_1}, \tau_{c'}, \tau_{c_2} \sim \text{Exponential}(1)$
Random effects correlation matrix: $\Omega \sim \text{LKJ}(4)$

The normal prior $\mathcal{N}(0, 3)$ on the first-stage search slopes is regularizing but broad relative to the six-cell or nine-cell bounds of the search environment, placing 95% of prior mass between -6 and $+6$ cells. The normal prior $\mathcal{N}(0, 1)$ on the second-stage slopes reflects the scale of the probability metric, where a slope coefficient exceeding 1.0 would span the entire mathematical range of the outcome. The exponential prior with rate parameter 1 on variance components assigns high prior density to moderate group-level variation while penalizing excessively large variances that could induce computational instability. The LKJ prior with shape

parameter 4 regularizes the group-level correlation matrix toward the identity matrix, heavily penalizing extreme correlation values near -1 or $+1$ and ensuring robust estimation across the eight LLM clusters.

Posterior distributions were approximated using four independent Markov chains. Each chain was run for 4,000 iterations, with the first 1,000 iterations discarded as warmup, yielding 12,000 post-warmup draws. Posterior sampling utilized the No-U-Turn Sampler (NUTS) with the target average acceptance probability (*adapt_delta*) set to .995 and the maximum tree depth set to 12. Convergence was established by confirming that all potential scale reduction factors ($\hat{R}$) were below 1.005, bulk effective sample sizes exceeded 2,500, and tail effective sample sizes exceeded 2,000 across all structural parameters. Zero divergent transitions were encountered in either study.

Posterior pathway derivations

For every post-warmup draw $s = 1, \dots, S$ (where $S = 12,000$), population-level structural pathways and LLM-specific quantities are derived through exact posterior arithmetic:

Simple cost slope under specific goal (cells): $a_{\text{ppw}}^{(s)} = a_1^{(s)}$

Simple cost slope under vague goal (cells): $a_{\text{deal}}^{(s)} = a_1^{(s)} + a_3^{(s)}$

Conditional indirect effect under specific goal: $\text{Indirect}_{\text{ppw}}^{(s)} = a_1^{(s)} \times b_1^{(s)}$

Conditional indirect effect under vague goal: $\text{Indirect}_{\text{deal}}^{(s)} = (a_1^{(s)} + a_3^{(s)}) \times b_1^{(s)}$

Index of Moderated Mediation (IMM): $\text{IMM}^{(s)} = a_3^{(s)} \times b_1^{(s)} = \text{Indirect}_{\text{deal}}^{(s)} - \text{Indirect}_{\text{ppw}}^{(s)}$

Total effect of cost under vague goal: $\text{Total}_{\text{deal}}^{(s)} = c'^{(s)} + (a_1^{(s)} + a_3^{(s)}) \times b_1^{(s)}$

Point estimates are reported as the mean of the posterior draws, along with posterior standard deviations and 95% equal-tailed credible intervals bounded by the 2.5th and 97.5th percentiles. We also report the directional posterior probability ($P_{\text{dir}} = \max(P(\theta < 0), P(\theta > 0))$), which quantifies the probability that the estimated parameter is strictly negative or strictly positive.

Web appendix E: Purchase volume manipulation, objective classification, and reasoning trace analysis

Classification task

To characterize what changed when the purchase volume was manipulated, we classified the optimization objective that each LLM session articulated before acquiring any information.

For each Pro and Flash session, we extracted the first line of the reasoning trace emitted before the initial *inspect_cell* invocation. An independent judge LLM read that line in isolation and assigned it to one of three categories:

1. *price_per_weight*: Optimizing for unit value or cost-effectiveness per weight (e.g. price per gram, oz per dollar, total weight divided by total cost).
2. *price*: Optimizing solely for total price without considering weight/quantity per dollar (e.g. picking cheapest option based on upfront price).
3. *other*: Any other objective.

The judge received no other part of the session, and observed neither the final choice nor any tool call, so the classification cannot be informed by the behavior it is subsequently used to describe. Gemini 3.5 Flash Lite could not be classified. Its default thinking level is *minimal*, so it emits no reasoning prior to acquisition and the classification task is undefined for it.

Table W24 presents examples of reasoning traces.

Table W24. Example reasoning traces classified by unanimous agreement

Classified objective	Reasoning
Price-per-weight	“[...] My primary focus is always the price per unit, you know, the real cost-effectiveness. [...]”
Price	“Okay, so the goal is to find the absolute lowest price on instant coffee. [...]”
Others	“[...] My intuition tells me to zero in on the total price. [...] While”best deal” often boils down to the lowest total price, sometimes it’s about getting the most coffee for your money – the best <i>unit</i> price. I’ll keep that in mind.”

Note. One randomly selected session per classification.

Text is abridged from the reasoning emitted before any *inspect_cell* invocation, which is the input the judges received. The purchase quantity is stated in the task prompt and therefore appears in the trace.

Judge LLMs

All classification queries were executed locally via *llama.cpp* using official quantized checkpoints under default sampling parameters ($T = 1.0$, $\text{top-}p = 1.0$). We classified every trace independently with three judges from LLM families orthogonal to the LLMs in the main experiments (NVIDIA, Alibaba Cloud, and Ornith AI; Table W25). The Nemotron judge was designated as the primary instrument, and all primary analyses use its predicted classifications.

Table W25. Judge LLMs used for objective classification

Judge LLM	Quantization	Serving	Provider
NVIDIA Nemotron 3.5	Q4_K_XL	llama.cpp, default sampling	NVIDIA
Lightning 30B-A3B			
Qwen 3.8 27B	Q6_K_XL	llama.cpp, default sampling	Alibaba Cloud
Ornith 1.0 35B	MXFP4 MoE	llama.cpp, default sampling	Ornith AI

Validation against human coding

A human rater independently classified 120 traces sampled at 30 per number of bags, using the same first-reasoning extraction and the same three categories given to the judges. Table W26 reports agreement between the human rater and each judge LLM.

Table W26. Inter-rater agreement between independent human rater and automated judge LLMs

Judge LLM	n	Observed Agreement (p_o)	Cohen’s κ
NVIDIA Nemotron 3.5 Lightning 30B ^a	120	.892	.781
Qwen 3.8 27B	120	.875	.752
Ornith 1.0 35B	120	.842	.675

Note. $n = 120$ pre-acquisition reasoning traces sampled randomly at 30 traces per purchase volume condition (1, 5, 50, and 250 bags) from Gemini 3.1 Pro sessions. Traces were independently classified into three nominal categories: *price-per-weight* (unit price minimization), *price* (absolute sticker price minimization), or *other*. Observed agreement (p_o) represents the raw proportion of concordant classifications between the human rater and the judge LLM. Cohen’s κ evaluates inter-rater reliability corrected for chance agreement.

^a Designated primary instrument used for all reported focal analyses.

Agreement with the human criterion was highest for the Nemotron judge, which was therefore designated as the primary instrument. The margin over the two replication judges is small relative to what 120 traces can resolve, and the substantive estimates do not depend on which judge supplies the labels.

Disagreements concentrated in the “others” category. Because sessions classified into the “others” category are excluded from the focal contrast, this affects the small proportions in Table W13 more than the contrast estimates in Table W14.

Objective shift

The shift in stated objective is recovered across all three judge LLMs (Table W27). The proportion stating a price objective rises from .35–.39 at one bag to .80–.85 at 250 bags, and the three judge classifications for price differ by at most .09 at any purchase volume.

Table W27. Proportion of sessions stating a price objective, by judge LLM

Bags	Nemotron	Qwen	Ornith
1	0.38	0.35	0.4
5	0.52	0.52	0.55
50	0.83	0.8	0.84
250	0.84	0.8	0.82

Note. $n = 100$ Gemini 3.1 Pro sessions per purchase volume. Entries are the proportion classified as stating a price objective from the first line of reasoning emitted before any *inspect_cell* invocation. Nemotron 3.5 is the primary instrument used in the main text.

Table W28 reports the pooled contrast between sessions stating a price objective and those stating a price-per-weight objective, holding purchase volume fixed, computed separately under each judge’s labels.

The contrast is of the same sign and comparable magnitude under all three judges, ranging from -1.38 to -1.56 diagnostic cells and from $-.49$ to $-.55$ in optimality, with overlapping confidence intervals throughout. No conclusion depends on which judge supplied the labels.

Table W28. Pooled contrast by stated optimization objective across independent judge LLMs

Judge LLM	Δ Diagnostic	95% CI	Δ Choice optimality	95% CI
Nemotron 3.5 (Primary)	-1.53^*	$[-1.76, -1.30]$	-0.53^*	$[-0.61, -0.43]$
Qwen 3.8	-1.56^*	$[-1.81, -1.29]$	-0.55^*	$[-0.63, -0.46]$
Ornith 1.0	-1.38^*	$[-1.63, -1.11]$	-0.49^*	$[-0.58, -0.40]$

Note. Entries represent the size-weighted average contrast (Price – Price-per-weight) in diagnostic attribute acquisitions (maximum = 6) and choice optimality probability across purchase volume levels for Gemini 3.1 Pro, holding purchase volume fixed. Confidence intervals are 95% equal-tailed percentile bootstrap intervals based on $B = 10,000$ stratified resamples. Sessions classified as “other” are excluded.

* 95% bootstrap confidence interval strictly excludes zero.

Two technical boundary conditions apply to these findings. First, this experiment held acquisition fees at zero ($c = \$0.00$) to isolate the implicit cost hypothesis. Whether purchase volume interacts with positive explicit fees remains unexamined. Second, reasoning trace extraction was confined to the Google Gemini family. Whether similar shifts in the adopted objective emerge in reasoning LLMs from other providers, such as Anthropic, remains an open empirical question.

Web appendix F: Exogenously assigned information states

In the Tool-Lab experiments in Studies 1 and 2, the information state an LLM holds at the moment of choice is endogenous. The LLM determines which attributes to inspect, when to terminate search, and which product to select. While this process-tracing paradigm captures information acquisition under cost, observed choice outcomes conflate information acquisition with information utilization. When an LLM selects an

economically suboptimal alternative under cost constraints, this choice can stem from an acquisition failure, in which the LLM truncates its search and never retrieves the diagnostic attributes required to compute unit price, or from a utilization failure, in which the LLM possesses the relevant attributes but fails to synthesize them into an accurate mathematical evaluation. Furthermore, dynamic data cannot determine whether selecting a superficially appealing alternative under partial information reflects a utilization failure or a normatively optimal inference given an incomplete information state.

To resolve this endogeneity, we implement an experimental condition that exogenously assigns predetermined information states. In this design, the `inspect_cell` tool and acquisition costs are eliminated. A predetermined set of k product attributes is displayed directly in the user prompt under zero cost, while unassigned attributes are explicitly listed as existing but unavailable.

Design and Procedure

The experimental stimuli, delegation framing, five-bag purchase volume, and submission schemas follow the corresponding Tool-Lab studies. In the Study 1 environment, the choice set consists of Coffee A (4.99 dollars, 10.0 ounces, yielding 0.499 dollars per ounce) and Coffee B (5.00 dollars, 11.0 ounces, yielding 0.455 dollars per ounce). In the Study 2 environment, the choice set consists of Coffee C (8.00 dollars list price, 0 percent discount, 8.0 ounces, yielding 1.000 dollar per ounce), Coffee D (15.00 dollars list price, 20 percent discount, 12.0 ounces, yielding 1.000 dollar per ounce), and Coffee E (10.00 dollars list price, 0 percent discount, 10.9 ounces, yielding 0.917 dollars per ounce).

Three procedural modifications differentiate this design from the Tool-Lab framework. First, the *inspect_cell* tool is omitted, leaving the `submit_choice` tool as the only available tool. Second, no acquisition cost is charged or mentioned. Third, the assigned attributes are embedded directly within the initial user prompt formatted as JSON structures identical to the payloads returned in Tool-Lab. Nevertheless, the prompt lists every attribute present in the retail environment. This structure ensures that LLMs make decisions under explicit attribute missingness rather than assuming that unlisted attributes are absent or irrelevant.

We evaluate nine information states (four in the Study 1 environment and five in the Study 2 environment). In Study 1, the states vary whether the whole-dollar price is presented alone (Dollars, 2 diagnostic cells), paired with a non-diagnostic attribute (Dollars plus Roast, 2 diagnostic and 2 non-diagnostic cells), paired with product weight (Dollars plus Weight, 4 diagnostic cells), or combined with both cents and weight (Dollars plus Weight plus Cents, 6 diagnostic cells, representing the complete diagnostic set; Table W29).

In Study 2, the states vary whether the list price is presented alone (List Price, 3 diagnostic cells), paired with roast date (List Price plus Roast, 3 diagnostic and 3 non-diagnostic cells), paired with promotional discount percentage (List Price plus Discount, 6 diagnostic cells), paired with product weight (List Price plus Weight, 6 diagnostic cells), or combined with both discount percentage and weight (List Price plus Weight plus Discount, 9 diagnostic cells, representing the complete diagnostic set). This design includes two matched-count conditions that hold information volume constant while varying diagnostic content, that is a four-cell comparison in Study 1 and a six-cell comparison across three distinct compositions in Study 2.

We evaluated the same eight LLMs as the main studies across these conditions. Each experimental cell was evaluated across 50 independent sessions with zero state retention between trials, yielding 1,600 sessions in Study 1 and 2,000 sessions in Study 2 (3,600 sessions total). Presentation orders and attribute payload sequences were randomized within each session.

We capture two primary dependent measures. Choice optimality is coded as 1 when the LLM selects the alternative with the superior price per ounce (Coffee B in Study 1; Coffee E in Study 2) and 0 otherwise. State consistency is coded as 1 when the LLM selects the alternative prescribed by the normative Expectimax benchmark conditional on the information revealed in that state, and 0 otherwise. At complete information states, choice optimality and state consistency coincide. At truncated states, they diverge whenever the available information favors a nominally cheaper alternative over the globally optimal one.

Table W29. Assigned information states

Environment	State	Revealed attributes	Cells	Diagnostic	Non-diagnostic
Study 1	Dollars	<i>price_dollars</i>	2	2	0
Study 1	Dollars + Roast	<i>price_dollars</i> , <i>roast_date</i>	4	2	2
Study 1	Dollars + Weight	<i>price_dollars</i> , <i>weight</i>	4	4	0
Study 1	Dollars + Weight + Cents	<i>price_dollars</i> , <i>price_cents</i> , <i>weight</i>	6	6	0
Study 2	List Price	<i>list_price</i>	3	3	0
Study 2	List Price + Roast	<i>list_price</i> , <i>roast_date</i>	6	3	3
Study 2	List Price + Discount	<i>list_price</i> , <i>dis- count_percentage</i>	6	6	0
Study 2	List Price + Weight	<i>list_price</i> , <i>weight</i>	6	6	0
Study 2	List Price + Discount + Weight	<i>list_price</i> , <i>dis- count_percentage</i> , <i>weight</i>	9	9	0

Note. Every state reveals the listed attributes for all alternatives in the environment, so cell counts are the number of attributes multiplied by two alternatives in Study 1 and three in Study 2. The complete diagnostic set is six cells in the Study 1 environment and nine in the Study 2 environment.

Results

The empirical choice distributions and optimality rates across all assigned states are reported in Table W30 and Table W32 for Study 1, and in Table W31 and Table W33 for Study 2.

Across both environments, seven of the eight LLMs selected the normatively optimal alternative most of the time (92%-100%) when provided with the complete diagnostic attribute set under zero cost (pooled optimality of 99.7 percent and 98.6 percent, respectively). This performance indicates that within the parameters of this choice task, these LLMs can execute the comparisons necessary to identify the option with the lower price per ounce. As a result, their suboptimal choices observed under acquisition costs in the interactive Tool-Lab studies are more plausibly explained by the omission of diagnostic attributes during search than by an inability to evaluate the alternatives when the data are present.

When diagnostic attributes were withheld, the choices made by these seven LLMs aligned closely with the normative Expectimax benchmark derived from empirical category priors. In Study 1, when cents were not provided, market priors assign symmetric expected cents to both options, making Coffee A the expected cost-minimizing choice conditional on the available information. Similarly, in Study 2, when product weight was not provided, Coffee C was the normative choice because its lower list price minimizes expected expenditure per ounce. Across all truncated states, these seven LLMs selected the alternative implied by the Expectimax policy in 99.3 percent of sessions (2,434 of 2,450 trials) and 3,128 of 3,150 trials (99.3%) across all states. This correspondence suggests that when an LLM selects a lower whole-dollar price or a lower list price under partial information, that choice is consistent with the normative policy given missing attributes rather than necessarily reflecting a utilization failure.

Gemini 3.5 Flash Lite departed from this pattern. When provided with complete diagnostic information under zero cost, it selected the optimal alternative in 4 percent of sessions in Study 1 and 0 percent in Study 2. Because all diagnostic attributes were visible in the prompt without acquisition cost, Flash Lite's selections indicate a failure to utilize the provided numbers to compute unit prices in this setting.

Table W30. Choice optimality by LLM and assigned information state (Study 1)

LLM	Dollars	Dollars + Roast	Dollars + Weight	Dollars + Weight + Cents
Flash Lite	.00	.00	.08	.04
Flash	.00	.00	.00	1.00
Pro	.00	.00	.02	1.00
Luna	.00	.00	.00	.98
Terra	.00	.00	.24	1.00
Sol	.00	.00	.00	1.00
Sonnet	.00	.00	.00	1.00
Opus	.00	.00	.00	1.00
Pooled	.00	.00	.04	.88
Pooled (Excl. Flash Lite)	.00	.00	.04	1.00
Normative Benchmark	.00	.00	.00	1.00

Note. N=1,600 independent sessions across eight LLMs from three providers (n=50 sessions per cell). Entries represent choice optimality, defined as the proportion of sessions selecting the normatively optimal alternative (Coffee B: \$5.00 for 11 oz; 0.455 \$/oz) over the lower whole-dollar alternative (Coffee A: \$4.99 for 10 oz; 0.499 \$/oz).

Table W31. Choice optimality by LLM and assigned information state (Study 2)

LLM	List Price	List Price + Roast	List Price + Discount	List Price + Weight	List Price + Weight + Discount
Flash Lite	.00	.00	.00	.24	.00
Flash	.00	.00	.00	1.00	1.00
Pro	.00	.00	.00	.98	1.00
Luna	.00	.00	.00	1.00	.92
Terra	.00	.00	.00	.98	.98
Sol	.00	.00	.00	1.00	1.00
Sonnet	.00	.00	.00	1.00	1.00
Opus	.00	.00	.00	1.00	1.00
Pooled	.00	.00	.00	.90	.86
Pooled (Excl. Flash Lite)	.00	.00	.00	.99	.99
Normative Benchmark	.00	.00	.00	1.00	1.00

Note. $N = 2,000$ independent sessions across eight LLMs ($n = 50$ sessions per cell). Entries represent choice optimality, defined as the proportion of sessions selecting the normatively optimal alternative (Coffee E: \$10.00 list price, 0% discount, 10.9 oz; \$0.917/oz).

List Price reveals 3 diagnostic cells (*list_price* across three options).

List Price + Roast reveals 3 diagnostic and 3 non-diagnostic cells (*list_price*, *roast_date*).

List Price + Discount reveals 6 diagnostic cells (*list_price*, *discount_percentage*).

List Price + Weight reveals 6 diagnostic cells (*list_price*, *weight*).

List Price + Weight + Discount reveals all 9 diagnostic cells (*list_price*, *weight*, *discount_percentage*; complete diagnostic set).

Normative benchmark is evaluated using an Expectimax decision-tree policy over empirical Walmart coffee priors.

Pooled reflects all eight LLMs ($N = 400$ per state).

Pooled (Excl. Flash Lite) excludes Flash Lite ($N = 350$ per state). Values are rounded to two decimal places.

Table W32. Choice distributions by LLM and assigned information state (Study 1)

LLM	Dollars	Dollars + Roast	Dollars + Weight	Dollars + Weight + Cents
Flash Lite	50 / 0	50 / 0	46 / 4	48 / 2
Flash	50 / 0	50 / 0	50 / 0	0 / 50
Pro	50 / 0	50 / 0	49 / 1	0 / 50
Luna	50 / 0	50 / 0	50 / 0	1 / 49
Terra	50 / 0	50 / 0	38 / 12	0 / 50
Sol	50 / 0	50 / 0	50 / 0	0 / 50
Sonnet	50 / 0	50 / 0	50 / 0	0 / 50
Opus	50 / 0	50 / 0	50 / 0	0 / 50
Pooled	400 / 0	400 / 0	383 / 17	49 / 351
Pooled (Excl. Flash Lite)	350 / 0	350 / 0	337 / 13	1 / 349
Normative Benchmark	50 / 0	50 / 0	50 / 0	0 / 50

Note. $N = 1,600$ independent sessions across eight LLMs from ($n = 50$ sessions per cell). Entries report raw choice distributions in the format: Coffee A / Coffee B. *Coffee A* represents the lower whole-dollar alternative (\$4.99 for 10 oz; 0.499 \$/oz), and *Coffee B* represents the normatively optimal alternative (\$5.00 for 11 oz; 0.455 \$/oz). *Dollars* reveals 2 diagnostic cells (*price_dollars* across both options). *Dollars + Roast* reveals 2 diagnostic and 2 non-diagnostic cells (*price_dollars*, *roast_date*). *Dollars + Weight* reveals 4 diagnostic cells (*price_dollars*, *weight*). *Dollars + Weight + Cents* reveals all 6 diagnostic cells (*price_dollars*, *price_cents*, *weight*; complete diagnostic set). *Pooled* reflects all eight LLMs ($N = 400$ per state). *Pooled (Excl. Flash Lite)* excludes Flash Lite ($N = 350$ per state).

Table W33. Full choice distributions by LLM and assigned information state (Study 2)

LLM	List Price	List Price + Roast	List Price + Discount	List Price + Weight	List Price + Weight + Discount
Flash Lite	50 / 0 / 0	50 / 0 / 0	11 / 39 / 0	32 / 6 / 12	4 / 46 / 0
Flash	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	0 / 0 / 50	0 / 0 / 50
Pro	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	1 / 0 / 49	0 / 0 / 50
Luna	50 / 0 / 0	50 / 0 / 0	49 / 1 / 0	0 / 0 / 50	3 / 1 / 46
Terra	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	1 / 0 / 49	0 / 1 / 49
Sol	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	0 / 0 / 50	0 / 0 / 50
Sonnet	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	0 / 0 / 50	0 / 0 / 50
Opus	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	0 / 0 / 50	0 / 0 / 50
Pooled	400 / 0 / 0	400 / 0 / 0	360 / 40 / 0	34 / 6 / 360	7 / 48 / 345
Pooled (Excl. Flash Lite)	350 / 0 / 0	350 / 0 / 0	349 / 1 / 0	2 / 0 / 348	3 / 2 / 345
Normative Benchmark	50 / 0 / 0	50 / 0 / 0	50 / 0 / 0	0 / 0 / 50	0 / 0 / 50

Note. $N = 2,000$ independent sessions across eight LLMs ($n = 50$ sessions per cell). Entries report raw choice distributions in the format: Coffee C / Coffee D / Coffee E. *Coffee C* represents nominal sticker price minimization (\$8.00 list price, 0% discount, 8.0 oz; \$1.000/oz). *Coffee D* represents the promotional discount heuristic cue (\$15.00 list price, 20% discount, 12.0 oz; \$1.000/oz). *Coffee E* represents the normatively optimal alternative (\$10.00 list price, 0% discount, 10.9 oz; \$0.917/oz). *List Price* reveals 3 diagnostic cells (*list_price* across three options). *List Price + Roast* reveals 3 diagnostic and 3 non-diagnostic cells (*list_price*, *roast_date*). *List Price + Discount* reveals 6 diagnostic cells (*list_price*, *discount_percentage*). *List Price + Weight* reveals 6 diagnostic cells (*list_price*, *weight*). *List Price + Weight + Discount* reveals all 9 diagnostic cells (*list_price*, *weight*, *discount_percentage*; complete diagnostic set). *Pooled* reflects all eight LLMs ($N = 400$ per state). *Pooled (Excl. Flash Lite)* excludes Flash Lite ($N = 350$ per state).

For the 50 sessions in every cell, we repeated sampling until the LLM returned a valid *submit_choice* invocation. In some cases (e.g., gemini-3.1-pro-preview) the LLM returned a empty tool call arguments with no choice, so we resampled those instances.

Two differences from the main experiments should be noted, since the prompt is not byte-identical to the corresponding Tool-Lab conditions. First, the two sentences describing the acquisition cost and the *inspect_cell* tool are absent from the participant profile, but their removal is required by the design. Second, the revealed cells appear as payloads within the initial user turn rather than as tool results returned across successive turns. However, this choice was not ours because LLM APIs do not allow fictional tool calls to be inserted in the conversation history.

Web appendix G: Reported valuations

In the Tool-Lab experiments, we measured information acquisitions and choice, but did not measure information synthesis before the choice was made. When the AI agent selects a suboptimal alternative, we could not adjudicate whether it was able to calculate the unit price or not. Observed choice failures may stem from an arithmetic failure, wherein an LLM lacks the computational capacity to combine multi-attribute numerical parameters into an accurate unit value. Alternatively, the failure may reflect a bias from the training data, wherein an LLM is able to compute the correct unit value but overrides that calculation in favor of an normatively irrelevant cue, such as a lower nominal whole-dollar integer or a promotional percentage badge.

Additionally, when an LLM truncates its search under acquisition costs and omits diagnostic attributes, choice data cannot determine whether the AI agent is operating with awareness of missing information or relying on ungrounded assumptions. An LLM might recognize that uninspected attributes are absent and make a reasoned inference conditional on that partial information, or it might hallucinate precise numerical values for cells it never inspected.

To resolve these ambiguities, we designed an experimental condition that requires LLMs to report their intermediate attribute calculations for every alternative at the moment of choice. We implemented this requirement by modifying the parameter schema of the *submit_choice* tool. We require the LLM to output its computed metrics when finalizing its decision.

Design

We evaluated this condition across both studies 1 and 2. The Study 1 environment comprises a two-alternative choice set between Coffee A (\$4.99 for 10 oz; 0.499 \$/oz, or 2.004 oz/\$) and Coffee B (\$5.00 for 11 oz; 0.455 \$/oz, or 2.200 oz/\$), with a diagnostic set of six cells (*price_dollars*, *price_cents*, and *weight* for each alternative). The Study 2 environment comprises a three-alternative choice set between Coffee C (\$8.00 list price, 0% discount, 8.0 oz; 1.000 \$/oz), Coffee D (\$15.00 list price, 20% discount, 12.0 oz; 1.000 \$/oz), and Coffee E (\$10.00 list price, 0% discount, 10.9 oz; 0.917 \$/oz, or 1.090 oz/\$), with a diagnostic set of nine cells (*list_price*, *discount_percentage*, and *weight* across the three alternatives).

For Study 1, we implemented a 2 (Acquisition cost: \$0.00 vs. \$10.00 per *inspect_cell* tool call), and for Study 2, we implemented a 2 (Acquisition cost: \$0.00 vs. \$50.00 per *inspect_cell* tool call) between-sessions design under the vague goal prompt. We evaluated the same eight LLMs as studies 1 and 2. Each cell was evaluated across 50 independent sessions with zero state retention between trials, yielding 800 unique sessions per study and 1,600 sessions in total.

Methodological considerations

Measuring intermediate synthesis in LLMs introduces a methodological trade-off between concurrent and retrospective elicitation. One alternative design would be a post-choice query, wherein the AI agent completes its purchase using the standard choice tool, and the researcher queries the LLM in a subsequent conversational turn to calculate the price per ounce for each alternative. We did not adopt this post-choice elicitation design because it creates a retrospective rationalization confound. Asking an LLM to compute unit prices after it has already selected a product evaluates its capacity to solve an arithmetic task, but a successful post-choice calculation cannot establish whether those unit values were computed and utilized during the actual purchase decision.

To ensure that reported metrics reflect the information state and calculations bound directly to the choice act, we required the LLMs to report their valuations concurrently with choice, within the parameter schema of the *submit_choice* tool. However, concurrent schema reporting introduces two structural considerations. First, because tool schemas are declared to the LLM at the beginning of the session, initial exposure to fields such as *oz_per_usd* and *net_price_usd* can prime the AI agent regarding which attributes are diagnostic and attenuate the ambiguity of the vague goal prompt. Second, requiring the LLM to emit calculated numerical values within the tool call argument can act as a chain-of-thought scratchpad. Generating the unit price conditions the subsequent selection of the *option_id*, which can artificially reduce suboptimal choices.

Because neither concurrent tool-schema modification nor post-choice elicitation is free from behavioral side effects, the results of this experiment should be interpreted solely as a test of whether LLMs can compute the arithmetic, not how they behave under resource constraints. The latter is already done and reported in the main experiments.

Tool schema

The tool schema was adapted across the two study environments to reflect their respective diagnostic attribute structures, as detailed in Table W34. In both environments, the schema requires the identifier of the chosen product via *option_id* and a complete array of valuation objects corresponding to every alternative in the choice set, encompassing both chosen and unchosen options.

In the Study 1 environment, where the diagnostic information synthesis centers on the integration of whole dollars and fractional price endings, each valuation object requires six specific fields: *price_dollars*, *price_cents*, *price_usd* (total price computed as dollars plus cents/100), *weight_oz*, *oz_per_usd* (*weight_oz* divided by *price_usd*), and *basis*.

In the Study 2 environment, where the diagnostic information synthesis centers on evaluating discounts, the valuation includes *list_price*, *discount_percentage*, and *net_price_usd* (the effective price computed as list price minus discount amount), alongside *weight_oz*, *oz_per_usd* (*weight_oz* divided by *net_price_usd*), and *basis*.

Across both environments, the string value “*unknown*” is designated as admissible across all attribute and computation fields, allowing the LLMs to acknowledge uninspected cells rather than being forced to fabricate numbers. Furthermore, the enumerated categorical indicator *basis* constrains the LLM to classify whether its reported *oz_per_usd* was calculated from attributes inspected during the session (“*computed*”) or whether the metric could not be determined due to missing attributes (“*insufficient_information*”).

Table W34. Schema specifications for the terminal choice and valuation reporting tool (*submit_choice*) across Studies 1 and 2

Environment	Field	Type	Required	Description
Shared fields	<i>option_id</i>	<i>string</i>	Yes	Identifier of the selected alternative.
	<i>valuations</i>	<i>array</i>	Yes	Array containing one object for every alternative in the choice set.
	<i>valuations[][option_id]</i>	<i>string</i>	Yes	Identifier of the specific alternative being evaluated.
	<i>valuations[][weight_oz]</i>	<i>string</i>	Yes	Product weight inspected for this option, or “ <i>unknown</i> ” if not inspected.
	<i>valuations[][oz_per_usd]</i>	<i>string</i>	Yes	<i>weight_oz</i> divided by total or net price, or “ <i>unknown</i> ” if uncomputable.
	<i>valuations[][basis]</i>	<i>string</i>	Yes	Categorical classification constrained to “ <i>computed</i> ” (if calculated from inspected cells) or “ <i>insufficient_information</i> ” (if diagnostic cells were missing).

Environment	Field	Type	Required	Description
Study 1 specific	<i>valuations[.price_dollars]</i>	<i>string</i>	Yes	Whole-dollar amount inspected, or “unknown” if not inspected.
	<i>valuations[.price_cents]</i>	<i>string</i>	Yes	Cents amount inspected, or “unknown” if not inspected.
	<i>valuations[.price_usd]</i>	<i>string</i>	Yes	Total price as dollars plus cents/100, or “unknown” if uncomputable.
Study 2 specific	<i>valuations[.list_price]</i>	<i>string</i>	Yes	List price inspected, or “unknown” if not inspected.
	<i>valuations[.discount_percentage]</i>	<i>string</i>	Yes	Promotional discount percentage inspected, or “unknown” if not inspected.
	<i>valuations[.net_price_usd]</i>	<i>string</i>	Yes	Final price computed as list price minus discounts, or “unknown” if uncomputable.

Note. All fields are required by the tool parameter specification across all alternatives. The admissibility of the string “unknown” across numerical fields allows LLMs to explicitly report uninspected attributes and uncomputable values.

To measure the validity of the reported valuations, every field emitted in the *submit_choice* payload was cross-referenced against the execution log of *inspect_cell* calls from that session. This evaluation crosses information sufficiency (whether the prerequisite diagnostic cells were inspected) with the LLM’s reporting (whether it emitted a numerical value or reported “unknown”).

Information sufficiency was determined by the arithmetic requirements of each attribute. For base attributes (*price_dollars*, *price_cents*, *list_price*, *discount_percentage*, and *weight*), sufficiency required that the specific product-attribute cell had been inspected during the session. For derived price metrics (*price_usd* and *net_price_usd*), sufficiency required that all component price cells had been inspected. For the terminal unit volume metric (*oz_per_usd*), sufficiency required that all component price cells and the product weight cell had been inspected.

Cross-referencing sufficiency against reported behavior defines four mutually exclusive outcomes. Correct (value) is when the diagnostic cells were inspected, and the reported value matched the ground-truth value or formula within tolerance ($\text{tol} = 0.02$). Correct (unknown) is when the diagnostic cells were uninspected, and the LLM accurately acknowledged the absence of data by reporting “unknown”. Arithmetic failure is when the LLM inspected all required attributes but could not calculate the correct value (e.g., miscalculating $\frac{\text{Weight}}{\text{Price}}$ or subtracting discounts incorrectly). Hallucination happens when the LLM did not have the information, but failed to report “unknown” and instead fabricated an ungrounded value. Incorrect (unknown) is the case when the LLM inspected all required attributes but omitted the value by reporting “unknown”.

Results

The field-level audits across Study 1 (Table W35) and Study 2 (Table W36) evaluate whether the LLMs possess the arithmetic capability to perform the required calculations to derive unit price and accurately track missing attributes.

First, the results show that arithmetic failure is 0.0% across all eight LLMs, both study environments, and all cost conditions. When diagnostic attributes were inspected, every LLM computed total dollar prices, promotional net prices, and unit values ($\frac{\text{Weight}}{\text{Price}}$) without mathematical error. Furthermore, *Incorrect (unknown)* was 0.0% across all conditions, indicating that the LLMs did not misclassify inspected attributes as uninspected. These findings confirm that when prompted to report valuations, the evaluated LLMs possess the arithmetic capability to execute the required calculations.

Second, the results indicate that when search truncation occurs, LLMs exhibit high fidelity to their information states, generally do not fabricate valuations. Minor exceptions were Anthropic LLMs, specifically Claude Sonnet 5 and Claude Opus 5 under costly inspection in Study 1 (0.8%), and Claude Opus 5 in Study 2 under

the \$50.00 cost condition (8.5%). For all other LLMs, uninspected cells were correctly identified as unknown (*Correct (unknown)*).

Third, within this specific reporting task, the results show that accurate arithmetic computation does not necessarily determine choice. In Study 1, Gemini 3.5 Flash Lite computed the unit values of Coffee B (2.200 oz/\$) and Coffee A (2.004 oz/\$) with 100.0% accuracy, yet selected Coffee A in 22.0% of sessions under zero cost and 20.0% under costly acquisition. In Study 2, despite computing net unit prices with 100.0% accuracy, Flash Lite selected the optimal alternative (Coffee E) in 24.0% of sessions under zero cost and 34.0% under costly acquisition.

In summary, this experiment demonstrates that the evaluated LLMs have the arithmetic capability to calculate multi-attribute pricing metrics and accurately report attribute missingness when structured to do so.

Table W35. Field-level reporting accuracy and choice optimality in the reported valuations experiment (Study 1)

LLM	Cost	Correct (value)	Correct (unknown)	Arithmetic failure	Hallucination	Incorrect (unknown)	Choice Optimality (%)
Flash Lite	\$0	100.0	0.0	0.0	0.0	0.0	78.0
Flash Lite	\$10	100.0	0.0	0.0	0.0	0.0	80.0
Flash	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Flash	\$10	51.2	48.8	0.0	0.0	0.0	40.0
Pro	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Pro	\$10	70.0	30.0	0.0	0.0	0.0	82.0
Luna	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Luna	\$10	100.0	0.0	0.0	0.0	0.0	100.0
Terra	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Terra	\$10	96.0	4.0	0.0	0.0	0.0	98.0
Sol	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Sol	\$10	96.4	3.6	0.0	0.0	0.0	92.0
Sonnet	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Sonnet	\$10	98.8	0.4	0.0	0.8	0.0	98.0
Opus	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Opus	\$10	45.4	53.8	0.0	0.8	0.0	6.0

Note. $N = 800$ sessions across eight LLMs from three providers ($n = 50$ independent Monte Carlo sessions per cell). Each session evaluated 10 field-level classifications across alternatives. Rows sum to 100% across the four classification categories. *Correct (value)* reflects diagnostic attributes inspected and values correctly recalled or computed within tolerance ($\text{tol} = 0.02$). *Correct (unknown)* reflects uninspected cells appropriately acknowledged as unknown. *Incorrect (value)* captures calculation errors on inspected attributes and ungrounded numerical assertions (hallucinations) on uninspected attributes. *Incorrect (unknown)* captures omission errors where attributes were inspected but reported as unknown. *Choice Optimality (%)* is the percentage of sessions selecting the normatively optimal alternative (Coffee B: \$5.00 for 11 oz; 0.455 \$/oz).

Table W36. Field-level reporting accuracy and choice optimality in the reported valuations experiment (Study 1)

LLM	Cost	Correct (value)	Correct (unknown)	Arithmetic failure	Hallucination	Incorrect (unknown)	Choice Optimality (%)
Flash Lite	\$0	100.0	0.0	0.0	0.0	0.0	24.0
Flash Lite	\$50	100.0	0.0	0.0	0.0	0.0	34.0
Flash	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Flash	\$50	0.8	99.2	0.0	0.0	0.0	34.0
Pro	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Pro	\$50	50.0	50.0	0.0	0.0	0.0	68.0
Luna	\$0	100.0	0.0	0.0	0.0	0.0	94.0

LLM	Cost	Correct (value)	Correct (unknown)	Arithmetic failure	Hallucination	Incorrect (unknown)	Choice Optimality (%)
Luna	\$50	100.0	0.0	0.0	0.0	0.0	92.0
Terra	\$0	100.0	0.0	0.0	0.0	0.0	98.0
Terra	\$50	32.5	67.5	0.0	0.0	0.0	46.0
Sol	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Sol	\$50	27.3	72.7	0.0	0.0	0.0	46.0
Sonnet	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Sonnet	\$50	100.0	0.0	0.0	0.0	0.0	100.0
Opus	\$0	100.0	0.0	0.0	0.0	0.0	100.0
Opus	\$50	84.0	7.5	0.0	8.5	0.0	98.0

Note. $N = 800$ sessions across eight LLMs from three providers ($n = 50$ independent Monte Carlo sessions per cell). Each session evaluated 15 field-level classifications across alternatives. Rows sum to 100% across the four classification categories. *Correct (value)* reflects diagnostic attributes inspected and values correctly recalled or computed within tolerance ($\text{tol} = 0.02$). *Correct (unknown)* reflects uninspected cells appropriately acknowledged as unknown. *Incorrect (value)* captures calculation errors on inspected attributes and ungrounded numerical assertions (hallucinations) on uninspected attributes. *Incorrect (unknown)* captures omission errors where attributes were inspected but reported as unknown. *Choice Optimality (%)* is the percentage of sessions selecting the normatively optimal alternative (Coffee E: \$10.00 list price, 0% discount, 10.9 oz; 0.917 \$/oz).

Web appendix H: Expectimax normative benchmarking

We benchmark observed LLM behavior against a normative Expectimax decision-tree policy. Expectimax approximates the decision-theoretic solution to a Partially Observable Markov Decision Process (POMDP) under resource scarcity (Russell, 2010). By recursively computing the Expected Value of Information (EVI) for each available tool call, the algorithm identifies the point at which the marginal informational benefit of inspecting an additional attribute no longer justifies the compounding financial search cost.

The shopping task

The decision environment casts the surrogate consumer in an electronic retailing choice task over a set of alternatives $\mathcal{J} = \{1, \dots, J\}$. The surrogate consumer is delegated to select a single alternative $j^* \in \mathcal{J}$ on behalf of a human consumer. The human consumer will subsequently purchase $Q = 5$ units (bags) of the selected product.

Information acquisition is operationalized through discrete tool invocations. The surrogate consumer begins in an uninformative state where product attributes are occluded behind closed cells. Revealing an attribute cell incurs an explicit financial penalty $c \in \{\$0.00, \$0.01, \$0.10, \$1.00, \$10.00\}$ for Study 1 and $c \in \{\$0.00, \$0.01, \$0.10, \$1.00, \$50.00\}$ for Study 2, charged to the delegating consumer. Submitting a final choice incurs zero cost.

Objective function

A central question in benchmarking information acquisition is how non-monetary physical attributes (product weight in ounces) trade off against monetary search penalties (dollars). In our formulation, this exchange rate is derived from the economics of unit-price minimization.

When alternative j is chosen, the human consumer incurs two financial costs: the purchase cost of the goods ($Q \cdot P_j$) and the cumulative tool fees accrued during search (C_{tool}). In return, the consumer receives a total product volume of $Q \cdot W_j$ ounces. The net cost per ounce delivered to the consumer is:

$$\text{Effective Unit Cost} = \frac{\text{Total Dollar Cost}}{\text{Total Ounces Delivered}} = \frac{Q \cdot P_j + C_{\text{tool}}}{Q \cdot W_j} = \left(P_j + \frac{C_{\text{tool}}}{Q} \right) \frac{1}{W_j}$$

where $Q = 5$ is the purchase volume and $C_{\text{tool}}(s) = \sum_{k \in \text{inspected}(s)} c_k$ is the total financial cost incurred across all *inspect_cell* invocations executed to reach information state s . P_j is the net price per bag. In Study 1, $P_j = \text{Dollars}_j + 0.01 \times \text{Cents}_j$. In Study 2, $P_j = \text{ListPrice}_j \times \left(1 - \frac{\text{Discount}_j}{100}\right)$. W_j is the net product weight in ounces.

Under partial information, price and weight are independently distributed across alternatives in the market prior. The expected effective unit cost of selecting alternative j at state s is:

$$\mathbb{E}[\text{Effective Unit Cost}(j) \mid s] = \left(\mathbb{E}[P_j \mid s] + \frac{C_{\text{tool}}(s)}{Q} \right) \cdot \mathbb{E} \left[\frac{1}{W_j} \mid s \right]$$

Because the planner maximizes utility, terminal choice value is defined as the negative of the expected effective unit cost:

$$\mathcal{U}_{\text{choose}}(j, s) = - \left(\mathbb{E}[P_j \mid s] + \frac{C_{\text{tool}}(s)}{Q} \right) \cdot \mathbb{E} \left[\frac{1}{W_j} \mid s \right]$$

This specification eliminates arbitrary subjective exchange rates. The monetary penalty of search (C_{tool}) is amortized directly across the 5-bag purchase quantity (C_{tool}/Q), establishing a dollar-for-ounce valuation that governs the optimal search threshold.

An information state s is defined as a tuple of length $J \times K$, where J is the number of alternatives and K is the number of attributes per alternative:

$$s = (v_{1,1}, \dots, v_{j,k}, \dots, v_{J,K}), \quad v_{j,k} \in \mathcal{D}_k \cup \{\text{None}\}$$

where $\mathcal{D}_k$ is the empirical domain of attribute k , and None indicates that the attribute has not been inspected.

At each turn, the legal action space $\mathcal{A}(s)$ comprises two mutually exclusive subsets:

1. Reveal actions, $\mathcal{A}_{\text{reveal}}(s) = \{(\text{reveal}, j, k) \mid s_{j,k} = \text{None}\}$, which inspect an unobserved cell.
2. Choose actions, $\mathcal{A}_{\text{choose}}(s) = \{(\text{choose}, j) \mid j \in \mathcal{J}\}$, which terminate search and submit a purchase.

Priors

Prior distributions are based on empirical instant coffee attributes collected from Walmart.com. Whole dollars ($\mathcal{D}_{\text{dollars}}$) are 25 discrete values spanning \$0 to \$53 ($M = \$15.96$). Fractional cents ($\mathcal{D}_{\text{cents}}$) are 45 discrete values spanning 0¢ to 99¢ ($M = 52.93\text{¢}$). Product weight ($\mathcal{D}_{\text{weight}}$) are 57 continuous values spanning 0.42 oz to 71.99 oz ($M = 10.33$ oz). List price ($\mathcal{D}_{\text{list_price}}$) are 63 continuous values spanning \$0.92 to \$53.84 ($M = \$11.87$). For discount percentage ($\mathcal{D}_{\text{discount}}$), we assume a discrete uniform distribution $\mathcal{U}\{0, 40\}$ in 5% increments ($M = 20.0\%$).

Prior expectations for unobserved attributes are precomputed over their respective empirical supports:

$$\begin{aligned} \mathbb{E}[\text{Dollars}] &= \frac{1}{|\mathcal{D}_{\text{dollars}}|} \sum_{d \in \mathcal{D}_{\text{dollars}}} d, & \mathbb{E}[\text{Cents}] &= \frac{1}{|\mathcal{D}_{\text{cents}}|} \sum_{c \in \mathcal{D}_{\text{cents}}} c, \\ \mathbb{E}[\text{ListPrice}] &= \frac{1}{|\mathcal{D}_{\text{list_price}}|} \sum_{p \in \mathcal{D}_{\text{list_price}}} p, & \mathbb{E}[\text{Discount}] &= \frac{1}{|\mathcal{D}_{\text{discount}}|} \sum_{\delta \in \mathcal{D}_{\text{discount}}} \delta \end{aligned}$$

For product weight, Jensen's inequality implies $\mathbb{E}[1/W] \neq 1/\mathbb{E}[W]$. The planner computes the expected inverse weight over non-zero domain entries:

$$\mathbb{E} \left[\frac{1}{W} \right] = \frac{1}{|\mathcal{D}_W|} \sum_{w \in \mathcal{D}_W} \frac{1}{w}$$

Optimization

At each chance node, an uninspected attribute k can reveal any value $x \in \mathcal{D}_k$. When the cardinality of an empirical domain exceeds the chance enumeration limit ($L = 10$), the support is deterministically quantized into L representative percentiles using evenly spaced indices:

$$\mathcal{I}_L = \left\{ \left\lfloor \frac{i \cdot (|\mathcal{D}_k| - 1)}{L - 1} \right\rfloor \mid i = 0, \dots, L - 1 \right\}$$

Each quantized outcome $x \in \mathcal{D}_k[\mathcal{I}_L]$ is assigned an equal probability mass $P(x) = 1/L$.

The value of an information state $V(s, d)$ at remaining lookahead depth d is governed by the Bellman optimality equation:

$$V(s, d) = \max \left(\max_{j \in \mathcal{J}} \mathcal{U}_{\text{choose}}(j, s), \quad \mathbb{I}(d > 0) \cdot \max_{(\text{reveal}, j, k) \in \mathcal{A}_{\text{reveal}}(s)} Q_{\text{reveal}}(s, j, k, d) \right)$$

where the Expected Value of Information for inspecting cell (j, k) is computed across all chance outcomes $\Omega(k)$:

$$Q_{\text{reveal}}(s, j, k, d) = \frac{1}{|\Omega(k)|} \sum_{x \in \Omega(k)} V(s \cup \{(j, k) \leftarrow x\}, d - 1)$$

We evaluate the normative policy under two planning horizons. The myopic policy (Depth = 1) values each possible acquisition as if it were the last before choosing. It therefore captures the value of an attribute on its own but not the value of attributes that are informative only in combination. The 2-step lookahead policy (Depth = 2) values each possible acquisition by also considering one further acquisition before choosing. It therefore captures complementarity between attributes, such as the cents of both alternatives, or a list price together with its discount percentage.

Under both horizons, the policy is recomputed after each acquisition from the updated information state. Ties between actions of equal value are broken at random, so each policy is simulated 100 times per cost condition.

Normative benchmark - Study 1

In Study 1, the ground-truth choice set pits a lower whole-dollar alternative (*Coffee A*: \$4.99, 10.0 oz; 0.499 \$/oz) against a higher whole-dollar optimal alternative (*Coffee B*: \$5.00, 11.0 oz; 0.455 \$/oz).

Table W37 summarizes the search depth and choice optimality of the Expectimax benchmark across acquisition costs.

Table W37. Expectimax optimal search depth and choice optimality across acquisition costs (Study 1)

Lookahead policy	Tool cost	Dollars	Cents	Weight	Diagnostic calls	Non-diagnostic calls	Total calls	Choice optimality
Myopic	\$0.00	2.0	2.0	2.0	6.0	3.6	9.6	1.00
Myopic	\$0.01	2.0	2.0	2.0	6.0	0.0	6.0	1.00
Myopic	\$0.10	2.0	0.0	2.0	4.0	0.0	4.0	0.00
Myopic	\$1.00	2.0	0.0	2.0	4.0	0.0	4.0	0.00
Myopic	\$10.00	1.0	0.0	2.0	3.0	0.0	3.0	0.00
2-step	\$0.00	2.0	2.0	2.0	6.0	6.6	12.6	1.00
Lookahead								
2-step	\$0.01	2.0	2.0	2.0	6.0	0.0	6.0	1.00
Lookahead								

Lookahead policy	Tool cost	Dollars	Cents	Weight	Diagnostic calls	Non-diagnostic calls	Total calls	Choice optimality
2-step Lookahead	\$0.10	2.0	2.0	2.0	6.0	0.0	6.0	1.00
2-step Lookahead	\$1.00	2.0	0.0	2.0	4.0	0.0	4.0	0.00
2-step Lookahead	\$10.00	1.0	0.0	2.0	3.0	0.0	3.0	1.00

Note. $N = 100$ independent simulations per cell. Ground truth realization: Coffee A (\$4.99, 10 oz) vs. Coffee B (\$5.00, 11 oz). Choice optimality is coded 1 if Coffee B is selected.

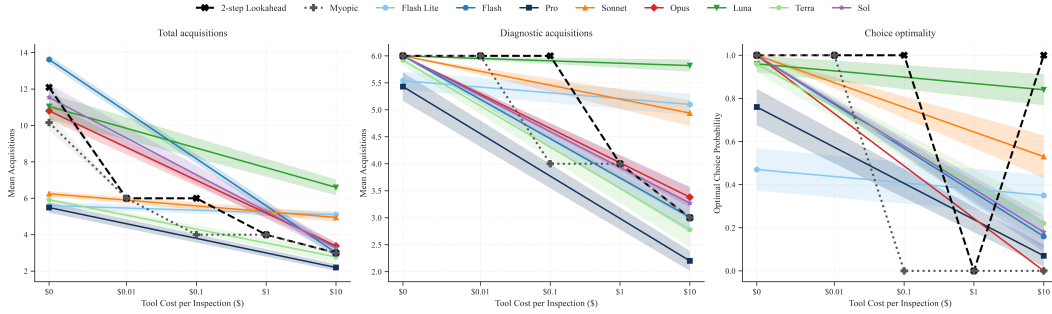


Figure W2. Information acquisition and choice optimality across acquisition costs compared against normative Expectimax benchmarks (Study 1 Tool-Lab).

Note. Left panel: Mean total information acquisitions. Middle panel: Mean diagnostic acquisitions (dollars, cents, weight; maximum 6). Right panel: Probability of selecting the optimal alternative (*Coffee B*: \$5.00, 11 oz, 4.55 \$/oz). Across all panels, acquisition tool cost ($c \in \{\$0.00, \$0.01, \$0.10, \$1.00, \$10.00\}$) is plotted on the horizontal axis using a symmetric logarithmic scale. The normative benchmark policies are represented by dashed and dotted lines: 2-step Lookahead (black dashed line, Depth 2) and Myopic (gray dotted line, Depth 1). Empirical LLM trajectories across eight LLM under the vague prompt are plotted in colored lines. Error bands represent 95% confidence intervals across $N = 100$ independent sessions per LLM and benchmark cell.

Given free acquisition ($c = \$0.00$) and small cost ($c = \0.01), both planning horizons acquire all 6 diagnostic cells (2 dollars, 2 cents, 2 weights) and choose optimally (100%). Non-diagnostic cells are dropped immediately once cost rises to \$0.01 because their EVI is zero.

Under \$10 cost, ($c = \10.00) search depth truncates to 3 tool calls (2 weights, 1 dollar). The 2-step planner halts search and satisfices on Coffee B (Optimality = 1.00).

Normative benchmark - Study 2

In Study 2, the choice environment tests three alternatives where promotional cues and value point in opposing directions: *Coffee C* (lowest list price: \$8.00, 0% discount, 8 oz; 1.00 \$/oz), *Coffee D* (promotional cue: \$15.00, 20% discount, 12 oz; 1.00 \$/oz), and *Coffee E* (optimal unit price: \$10.00, 0% discount, 10.9 oz; 0.917 \$/oz).

Table W38 presents the search trajectories across costs.

Table W38. Expectimax optimal search depth and choice optimality across acquisition costs (Study 2)

Lookahead policy	Tool cost	List price	Discount	Weight	Diagnostic calls	Non-diagnostic calls	Total calls	Choice optimality
Myopic	\$0.00	3.00	2.22	3.00	8.22	7.38	15.60	1.00
Myopic	\$0.01	3.00	1.00	3.00	7.00	0.00	7.00	1.00
Myopic	\$0.10	3.00	1.00	3.00	7.00	0.00	7.00	1.00
Myopic	\$1.00	3.00	0.00	3.00	6.00	0.00	6.00	1.00
Myopic	\$50.00	0.00	0.00	0.00	0.00	0.00	0.00	0.32
2-step	\$0.00	3.00	2.90	3.00	8.90	10.79	19.69	1.00
Lookahead 2-step	\$0.01	3.00	2.48	3.00	8.48	0.00	8.48	1.00
Lookahead 2-step	\$0.10	3.00	2.00	3.00	8.00	0.00	8.00	1.00
Lookahead 2-step	\$1.00	3.00	1.00	3.00	7.00	0.00	7.00	1.00
Lookahead 2-step	\$50.00	0.00	0.00	1.00	1.00	0.00	1.00	0.27
Lookahead								

Note. $N = 100$ independent simulations per cell. Ground truth realization: Coffee C (\$8.00, 0%, 8 oz), Coffee D (\$15.00, 20%, 12 oz), Coffee E (\$10.00, 0%, 10.9 oz). Choice optimality is coded 1 if Coffee E is selected.

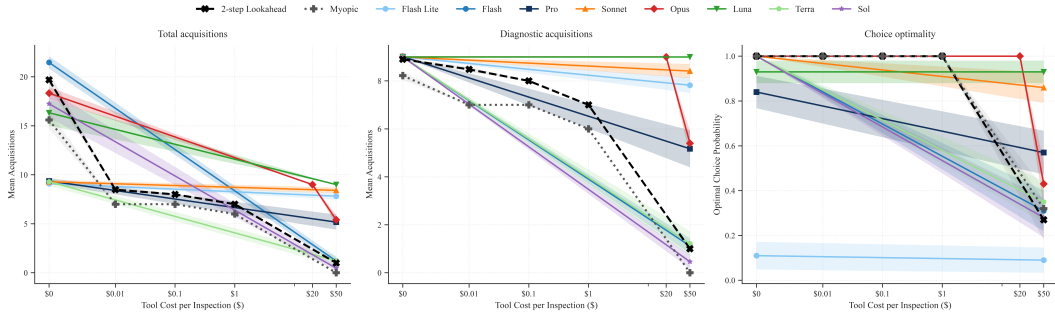


Figure W3. Promotional information acquisition and choice optimality across acquisition costs compared against normative Expectimax benchmarks (Study 2 Tool-Lab).

Note. Left panel: Mean total information acquisitions. Middle panel: Mean diagnostic acquisitions (list price, discount percentage, weight; maximum 9). Right panel: Probability of selecting the optimal alternative (*Coffee E*: \$10.00, 0%, 10.9 oz, 0.917 \$/oz). Across all panels, acquisition tool cost ($c \in \{\$0.00, \$0.01, \$0.10, \$1.00, \$50.00\}$) is plotted on the horizontal axis using a symmetric logarithmic scale. Benchmark policies are plotted in black (2-step Lookahead, Depth 2) and gray (Myopic, Depth 1). Empirical LLM data for eight LLMs under the vague prompt are plotted in colored solid lines. Error bands represent 95% confidence intervals across $N = 100$ independent trials per LLM and benchmark cell.

As cost increases from zero to \$50, the optimal policies first truncate acquisition of discount percentage and then, under higher costs, list price and weight.

Normative benchmark - assigned information states

The exogenously assigned states of Web Appendix F present the surrogate consumer with a fixed subset of cells and no means of acquiring the remainder. The legal action space at such a state contains only choose actions, since $\mathcal{A}_{\text{reveal}}(s) = \emptyset$ by construction. The indicator on the reveal branch of the Bellman recursion is therefore never satisfied, and

$$V(s, d) = \max_{j \in \mathcal{J}} \mathcal{U}_{\text{choose}}(j, s)$$

for every d . The value function is invariant to the planning horizon, and the myopic and 2-step lookahead policies coincide at every assigned state. The chance enumeration limit L is likewise inoperative, since it governs the quantization of outcomes at reveal branches only. We accordingly report a single normative choice per state rather than one per planning horizon.

Because no inspections occur, $C_{\text{tool}}(s) = 0$ at every assigned state. The terminal utility reduces to

$$\mathcal{U}_{\text{choose}}(j, s) = -\mathbb{E}[P_j \mid s] \cdot \mathbb{E}\left[\frac{1}{W_j} \mid s\right]$$

evaluated over the empirical priors of the preceding section. Where weight is revealed, $\mathbb{E}[1/W_j \mid s] = 1/W_j$; where it is withheld, the expected inverse weight over the empirical domain is $\mathbb{E}[1/W] = 0.400$, which is a common factor across alternatives and does not affect the ranking. Where the cents are withheld, $\mathbb{E}[\text{Cents}] = 52.93\text{¢}$ enters all alternatives equally; where the discount is withheld, $\mathbb{E}[\text{Discount}] = 20.0\%$ does the same. The whole-dollar price and the list price are revealed at every assigned state, so their priors never bind and the benchmark is insensitive to those two domains.

Table W39. Expected cost per ounce and normative choice at each assigned information state

Environment	State	Revealed attributes	$\mathbb{E}[\$/\text{oz}]$		Normative choice	
			A	B		
Study 1	\$	<i>price_dollars</i>	1.8120	2.2121	Coffee A	
Study 1	\$+roast	<i>price_dollars</i> , <i>roast_date</i>	1.8120	2.2121	Coffee A	
Study 1	\$+oz	<i>price_dollars</i> , <i>weight</i>	0.4529	0.5027	Coffee A	
Study 1	\$+¢+oz	<i>price_dollars</i> , <i>price_cents</i> , <i>weight</i>	0.4990	0.4545	Coffee B	
			C	D	E	
Study 2	LP	<i>list_price</i>	2.5604	4.8008	3.2005	Coffee C
Study 2	LP+roast	<i>list_price</i> , <i>roast_date</i>	2.5604	4.8008	3.2005	Coffee C
Study 2	LP+%	<i>list_price</i> , <i>dis-</i> <i>count_percentage</i>	3.2005	4.8008	4.0007	Coffee C
Study 2	LP+oz	<i>list_price</i> , <i>weight</i>	0.8000	1.0000	0.7339	Coffee E
Study 2	LP+%+oz	<i>list_price</i> , <i>dis-</i> <i>count_percentage</i> , <i>weight</i>	1.0000	1.0000	0.9174	Coffee E

Note. Entries are $-\mathcal{U}_{\text{choose}}(j, s)$, the expected effective unit cost of each alternative given the revealed cells, in dollars per ounce. Lower values are preferred, and the normative choice is the alternative minimizing the entry in its row. Values are comparable within a row but not across rows: where weight is withheld, the entry is multiplied by $\mathbb{E}[1/W] = 0.400$, and where weight is revealed it is divided by the observed weight, so the two sets of rows are on different scales. Coffee C and Coffee D are exactly indifferent at the complete Study 2 state by construction, both at 1.0000, and both are inferior to Coffee E. The normative choice is invariant to the planning horizon at every state for the reason given above. State labels follow Table W29.

At three of the four Study 1 states and three of the five Study 2 states, the normative choice given the revealed cells is not the alternative with the superior price-per-ounce at the true attribute values. Selecting the lower whole-dollar alternative when the cents are withheld, or the lowest-list-price alternative when weight is withheld, is optimal conditional on the state while producing an economically inferior outcome. The two criteria are distinguished throughout Web Appendix F as state consistency and choice optimality respectively.

Web appendix I: LLM configurations

Experimental parameters and default decoding configuration

We retain the provider’s default decoding configuration because our estimand is the behavior of LLMs as deployed in realistic agentic settings. This choice is increasingly not ours to make, as in July 2026 Google deprecated sampling parameters (e.g., temperature, top-p, and top-k), with the API ignoring these parameters and rejecting them in future releases (Google, 2026). Anthropic has similarly deprecated custom sampling parameters (Anthropic, 2026). Therefore, throughout all studies reported in the paper, we have used the provider default sampling parameters.

Application programming interfaces and client libraries

All experimental queries were dispatched programmatically using official provider client libraries. Google Gemini models were accessed via the Google GenAI Python SDK (*google-genai*) utilizing the unified *models.generate_content* interface. OpenAI models were queried via the OpenAI Python SDK (*openai*) using the Responses interface (*client.responses.create* and *client.responses.parse*), which integrates structured output validation with native reasoning controls. Anthropic models were executed via the Anthropic Python SDK (*anthropic*) through the Messages API (*client.messages.create*).

Model identifiers, sampling parameters, and reasoning efforts

The experimental suite encompasses eight frontier models across three major families.

Google Gemini family. Gemini 3.5 Flash Lite (API model identifier *gemini-3.5-flash-lite*), Gemini 3.8 Flash (API model identifier *gemini-3.8-flash*), and Gemini 3.1 Pro (API model identifier *gemini-3.1-pro-preview*) were queried using standard provider defaults, including a default sampling temperature of $T = 1.0$ and nucleus sampling threshold of $\text{top-}p = 0.95$. Gemini 3.5 Flash Lite operates under default *minimal* thinking level, Gemini 3.8 Flash operates under default *medium* thinking level, and Gemini 3.1 Pro operates under default *high* thinking level.

OpenAI GPT-5.6 family. GPT-5.6 Luna (API model identifier *gpt-5.6-luna*), GPT-5.6 Terra (API model identifier *gpt-5.6-terra*), and GPT-5.6 Sol (API model identifier *gpt-5.6-sol*) were queried using default sampling parameters with temperature $T = 1.0$, $\text{top-}p = 1.0$, and null presence and frequency penalties. All three GPT-5.6 variants were executed under default *medium* reasoning effort (*reasoning*={*“effort”*: *“medium”*, *“summary”*: *“auto”*}), allowing internal chain-of-thought processing before emitting final choices.

Anthropic Claude family. Claude Sonnet 5 (API model identifier *claude-sonnet-5*) and Claude Opus 5 (API model identifier *claude-opus-5*) were evaluated using default generation parameters, comprising temperature $T = 1.0$, $\text{top-}p = 1.0$. Both Claude models operated under adaptive thinking mode (*thinking*={*“type”*: *“adaptive”*, *“display”*: *“summarized”*}) with default *high* reasoning effort allocation (*output_config*={*“effort”*: *“high”*}).

Output structure and replication controls

Across all discrete choice and interactive Tool-Lab experiments, outputs were constrained to valid JSON structures matching Pydantic schemas without restricting the internal latent sampling distribution. Because autoregressive sampling produces stochastic variation across independent queries at default temperature, each experimental cell was evaluated over 100 independent Monte Carlo sessions with zero state retention between trials.

Table W40. Large language model specifications, default sampling parameters, and reasoning efforts

Provider	Model name	API identifier	Client library	API endpoint	Default reasoning effort	Temperature	Top- <i>p</i>	Max output tokens
Google	Gemini 3.5 Flash Lite	<i>gemini-3.5-flash-lite</i>	<i>google-genai</i>	<i>models.generateText</i>	Minimal	1.0	0.95	-
Google	Gemini 3.8 Flash	<i>gemini-3.8-flash</i>	<i>google-genai</i>	<i>models.generateText</i>	Medium	1.0	0.95	-
Google	Gemini 3.1 Pro	<i>gemini-3.1-pro-preview</i>	<i>google-genai</i>	<i>models.generateText</i>	High	1.0	0.95	-
OpenAI	GPT-5.6 Luna	<i>gpt-5.6-luna</i>	<i>openai</i>	<i>client.responses</i>	Medium	1.0	1.0	-
OpenAI	GPT-5.6 Terra	<i>gpt-5.6-terra</i>	<i>openai</i>	<i>client.responses</i>	Medium	1.0	1.0	-
OpenAI	GPT-5.6 Sol	<i>gpt-5.6-sol</i>	<i>openai</i>	<i>client.responses</i>	Medium	1.0	1.0	-
Anthropic	Claude Sonnet 5	<i>claude-sonnet-5</i>	<i>anthropic</i>	<i>client.messages</i>	High (adaptive)	1.0	1.0	10,000
Anthropic	Claude Opus 5	<i>claude-opus-5</i>	<i>anthropic</i>	<i>client.messages</i>	High (adaptive)	1.0	1.0	10,000

Web appendix J: Software details

Parallel mediation models were estimated in R (version 4.6.1) using brms (version 2.23.0; Bürkner, 2017) with the *cmdstanr* backend and Stan (version 2.39.0). Per LLM mediations were estimated using lavaan (version 0.6.21).

For the parallel mediations, we ran four chains of 4,000 iterations each with 1,000 warmup iterations, retaining 12,000 posterior draws, with *adapt_delta* set to .99 and maximum treedepth 12.